

Unit- II : Functions of Several variables

- Introduction
- How to find partial derivatives?
- Euler's Theorem for Homogeneous functions
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- Total derivative
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Introduction

- A real valued function of several variables is a function
- $f: D \rightarrow \mathbb{R}$, where the domain D is a subset of $\mathbb{R}^n$
- So for each $(x_1, x_2, x_3, \dots, x_n)$ in D , the value of f is a real number $f(x_1, x_2, x_3, \dots, x_n)$
- For example, the volume of a cylinder: $V = \pi r^2 h$
- i.e., $V = f(r, h)$ is a function of two variables.
- If f is defined by a formula, we usually take the domain D to be as large as possible.
- For example, if f is a function defined by $f(x, y) = 9 - \sin x + \cos(x^2 + y^2)$ we have a function of 2 variables defined for all $(x, y) \in \mathbb{R}^2$. So $D = \mathbb{R}^2$.
- However, if f is defined by, $f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$
- Then this is a function of 3 variables, defined for all $(x, y, z) \in \mathbb{R}^3$ except for $(x, y, z) = (0, 0, 0)$.

Introduction

Consider the cylinder of radius r and height h .

The volume V of a cylinder is given by: $V = \pi r^2 h$

V is a function of two variables radius r and height h

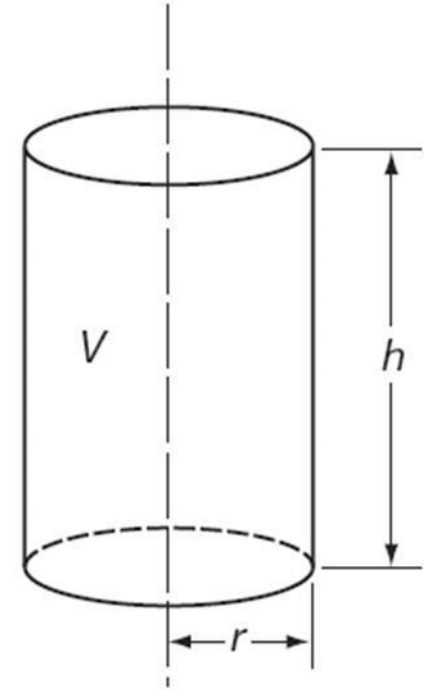
If r is kept constant and h increases then V increases.

We can find the rate of change of V with respect to h by differentiating V with respect to h , keeping r constant:

$$\left[\frac{dV}{dh} \right]_{r \text{ constant}} = \pi r^2$$

We write this as $\frac{\partial V}{\partial h} = \pi r^2$ and we read it as *dabba V* by *dabba h* or *del V* by *del h*.

This is called the **first partial derivative** of V with respect to h .



Similarly, if h is kept constant and r increases then V increases. We can then find the rate of change of V by differentiating with respect to r keeping h constant:

$$\left[\frac{dV}{dr} \right]_{h \text{ constant}} = 2\pi r h$$

We write this as $\frac{\partial V}{\partial r} = 2\pi r h$

We read it as *dabba* V by *dabba* r or *del* V by *del* r

This is called the **first partial derivative** of V with respect to r .

How to find partial derivatives?

If $z = f(x, y)$ is a function of two real variables x and y , then partial derivative of z w.r.to x is denoted by

$$\frac{\partial z}{\partial x} \text{ or } z_x \text{ or } \frac{\partial f}{\partial x} \text{ or } f_x$$

and **is the ordinary derivative of z w.r.to x by treating variable y as constant.**

Similarly partial derivative of z w.r.to y is denoted by

$$\frac{\partial z}{\partial y} \text{ or } z_y \text{ or } \frac{\partial f}{\partial y} \text{ or } f_y$$

and **is the ordinary derivative of z w.r.to y by treating variable x as constant.**

Note : (1) $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ are called as first order partial derivatives of z .

$$(2) \frac{\partial z}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

$$\frac{\partial z}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k}$$

(3) If $f(x, y, z)$ is a function of three variables x, y and z then partial derivative of f w.r.to any single variable is obtained by treating remaining two variables constant.

(4) Standard rules for derivatives of sum, difference, product and quotient are also applicable for partial derivatives.

1. Find the values of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at the point $(4, -5)$ if $f(x, y) = x^2 + 3xy + y - 1$

Solution: To find $\frac{\partial f}{\partial x}$, treat y as a constant and differentiate with respect to x .

$$\therefore \frac{\partial f}{\partial x} = 2x + 3y$$

To find $\frac{\partial f}{\partial y}$, treat x as a constant and differentiate with respect to y .

$$\therefore \frac{\partial f}{\partial y} = 3x + 1$$

$$\Rightarrow \left(\frac{\partial f}{\partial x} \right)_{(4, -5)} = 2(4) + 3(-5) = -7$$

$$\left(\frac{\partial f}{\partial y} \right)_{(4, -5)} = 3(4) + 1 = 13$$

2. Find the values of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ where $f(x, y) = y \sin(xy)$.

Solution: $f(x, y) = y \sin(xy) \dots \dots \dots (1)$

Differentiate (1) w.r.to x by treating y as a constant

$$\therefore \frac{\partial f}{\partial x} = y \cos(xy) \times y = y^2 \cos(xy)$$

Differentiate (1) w.r.to y by treating x as a constant

$$\begin{aligned} \therefore \frac{\partial f}{\partial y} &= \sin(xy) + y \cos(xy) \times x \\ &= \sin(xy) + xy \cos(xy) \end{aligned}$$

3. Find f_x and f_y if $f(x, y) = \frac{2y}{y + \cos x}$

Solution: $f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \left(\frac{2y}{y + \cos x} \right)$

$$= \frac{(y + \cos x) \frac{\partial}{\partial x} (2y) - (2y) \frac{\partial}{\partial x} (y + \cos x)}{(y + \cos x)^2}$$
$$= \frac{(y + \cos x)(0) - (2y)(0 - \sin x)}{(y + \cos x)^2}$$
$$= \frac{0 - (2y)(-\sin x)}{(y + \cos x)^2}$$
$$= \frac{2y \sin x}{(y + \cos x)^2}$$

$$\begin{aligned}
 f_y &= \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \left(\frac{2y}{y + \cos x} \right) \\
 &= \frac{(y + \cos x) \frac{\partial}{\partial y} (2y) - (2y) \frac{\partial}{\partial y} (y + \cos x)}{(y + \cos x)^2} \\
 &= \frac{(y + \cos x)(2) - (2y)(1 + 0)}{(y + \cos x)^2} \\
 &= \frac{2(y + \cos x) - 2y}{(y + \cos x)^2} \\
 &= \frac{2y + 2\cos x - 2y}{(y + \cos x)^2} \\
 &= \frac{2\cos x}{(y + \cos x)^2}
 \end{aligned}$$

Second-Order Partial Derivatives

By differentiating a function $z = f(x, y)$ twice, we get its second-order derivatives. These derivatives are usually denoted by:

$$\frac{\partial^2 f}{\partial x^2} \text{ or } f_{xx} \text{ or } \frac{\partial^2 z}{\partial x^2} \text{ or } z_{xx}$$

$$\frac{\partial^2 f}{\partial y^2} \text{ or } f_{yy} \text{ or } \frac{\partial^2 z}{\partial y^2} \text{ or } z_{yy}$$

$$\frac{\partial^2 f}{\partial x \partial y} \text{ or } f_{xy} \text{ or } \frac{\partial^2 z}{\partial x \partial y} \text{ or } z_{xy}$$

$$\frac{\partial^2 f}{\partial y \partial x} \text{ or } f_{yx} \text{ or } \frac{\partial^2 z}{\partial y \partial x} \text{ or } z_{yx}$$

The defining equations are:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$$

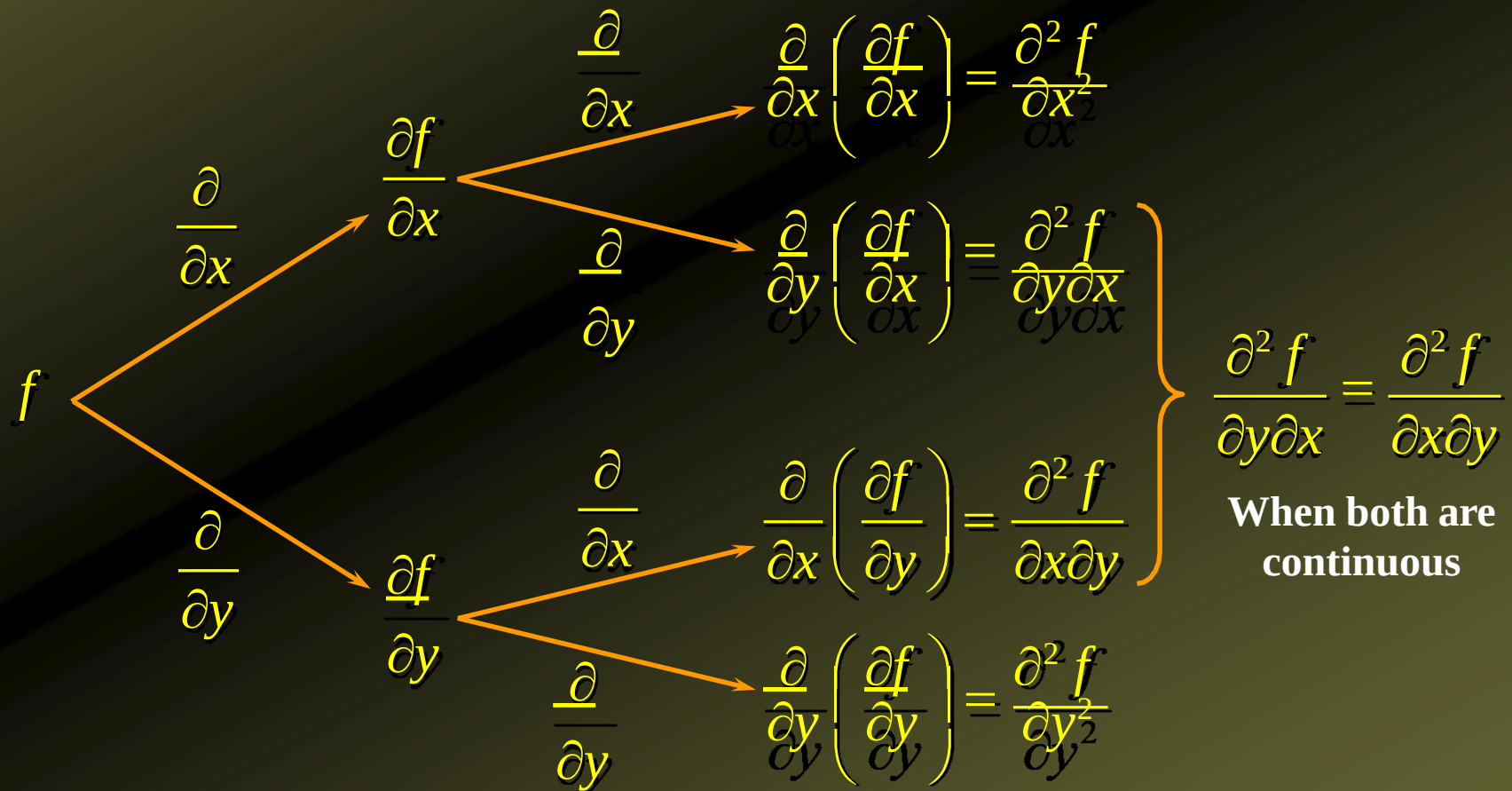
$$f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)$$

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \quad \longrightarrow \quad \text{Differentiate first w.r.t } y \text{ and then w.r.t. } x$$

$$f_{yx} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \quad \longrightarrow \quad \text{Differentiate first w.r.t } x \text{ and then w.r.t. } y$$

Second Order Partial Derivatives

- ◆ Thus, **four second-order partial derivatives** can be obtained of a function of **two variables**:



1. If $f(x, y) = x \cos y + y e^x$, then find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$, $\frac{\partial^2 f}{\partial x \partial y}$, $\frac{\partial^2 f}{\partial y \partial x}$

Solution:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x \cos y + y e^x) = \cos y + y e^x$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x \cos y + y e^x) = -x \sin y + e^x$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (\cos y + y e^x) = 0 + y e^x = y e^x$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (-x \sin y + e^x) = -x \cos y + 0 = -x \cos y$$

$$\begin{aligned}\frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (-x \sin y + e^x) \\ &= -\sin y + e^x\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (\cos y + y e^x) \\ &= -\sin y + e^x\end{aligned}$$

2. If $u = \tan^{-1} \left(\frac{y}{x} \right)$ then find $\frac{\partial^2 u}{\partial x^2}$, $\frac{\partial^2 u}{\partial y^2}$, $\frac{\partial^2 u}{\partial x \partial y}$ and $\frac{\partial^2 u}{\partial y \partial x}$

Solution: $\frac{\partial u}{\partial x} = \frac{1}{1 + \left(\frac{y}{x} \right)^2} \left(-\frac{y}{x^2} \right) = -\frac{y}{x^2 + y^2}$

$$\frac{\partial u}{\partial y} = \frac{1}{1 + \left(\frac{y}{x} \right)^2} \left(\frac{1}{x} \right) = \frac{x}{x^2 + y^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(-\frac{y}{x^2 + y^2} \right)$$

$$= -\frac{(x^2 + y^2) \frac{\partial}{\partial x} (y) - y \frac{\partial}{\partial x} (x^2 + y^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial x^2} = -\frac{(x^2+y^2)(0) - y(2x)}{(x^2+y^2)^2} = \frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{x}{x^2+y^2} \right)$$

$$= \frac{(x^2+y^2) \frac{\partial}{\partial y} (x) - x \frac{\partial}{\partial y} (x^2+y^2)}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)(0) - x(2y)}{(x^2+y^2)^2} = -\frac{2xy}{(x^2+y^2)^2}$$

$$\begin{aligned}
\frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2} \right) \\
&= \frac{(x^2 + y^2) \frac{\partial}{\partial x} (x) - x \frac{\partial}{\partial x} (x^2 + y^2)}{(x^2 + y^2)^2} \\
&= \frac{(x^2 + y^2)(1) - x(2x)}{(x^2 + y^2)^2} \\
&= \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 u}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial y} \left(\frac{-y}{x^2 + y^2} \right) \\
&= - \frac{(x^2 + y^2) \frac{\partial}{\partial y} (y) - y \frac{\partial}{\partial y} (x^2 + y^2)}{(x^2 + y^2)^2} \\
&= - \frac{(x^2 + y^2)(1) - y(2y)}{(x^2 + y^2)^2} \\
&= - \frac{(x^2 + y^2 - 2y^2)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}
\end{aligned}$$

3. Verify $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ for i) $u = \log(x^2 + y^2)$ ii) $u = x^y + y^x$

Solution: i) $u = \log(x^2 + y^2)$

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{2y}{x^2 + y^2} \right) = 2y \frac{\partial}{\partial x} \left(\frac{1}{x^2 + y^2} \right)$$

$$= 2y \left[\frac{-1}{(x^2 + y^2)^2} \right] (2x) \quad \left[\because \frac{d}{dx} \left(\frac{1}{f(x)} \right) = \frac{-1}{(f(x))^2} f'(x) \right]$$

$$= \frac{-4xy}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial y} \left(\frac{2x}{x^2 + y^2} \right) = 2x \frac{\partial}{\partial y} \left(\frac{1}{x^2 + y^2} \right)$$

$$= 2x \left[\frac{-1}{(x^2 + y^2)^2} \right] (-2y) \quad \left[\because \frac{d}{dy} \left(\frac{1}{f(y)} \right) = \frac{-1}{(f(y))^2} f'(y) \right]$$

$$= \frac{-4xy}{(x^2 + y^2)^2}$$

$$\therefore \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

$$\text{ii) } u = x^y + y^x$$

$$\frac{\partial u}{\partial x} = yx^{y-1} + y^x \log y$$

$$\frac{\partial u}{\partial y} = x^y \log x + xy^{x-1}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} (x^y \log x + xy^{x-1})$$

$$= \frac{\partial}{\partial x} (x^y \log x) + \frac{\partial}{\partial x} (xy^{x-1})$$

$$= yx^{y-1} \log x + x^y \cdot \frac{1}{x} + y^{x-1} + x y^{x-1} \log y$$

$$= x^{y-1} + y^{x-1} + yx^{y-1} \log x + x y^{x-1} \log y$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial y} (y x^{y-1} + y^x \log y)$$

$$= \frac{\partial}{\partial y} (y x^{y-1}) + \frac{\partial}{\partial y} (y^x \log y)$$

$$= x^{y-1} + y x^{y-1} \log x + x y^{x-1} \log y + y^x \cdot \frac{1}{y}$$

$$= x^{y-1} + y^{x-1} + y x^{y-1} \log x + x y^{x-1} \log y$$

$$\therefore \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

4. If $u = \log(\tan x + \tan y + \tan z)$ then show that

$$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$$

Solution: $u = \log(\tan x + \tan y + \tan z)$

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 x$$

$$\frac{\partial u}{\partial y} = \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 y$$

$$\frac{\partial u}{\partial z} = \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 z$$

$$\sin 2x \frac{\partial u}{\partial x} = \sin 2x \frac{\sec^2 x}{\tan x + \tan y + \tan z}$$

$$\begin{aligned}\sin 2x \frac{\partial u}{\partial x} &= \frac{2 \sin x \cos x \sec^2 x}{\tan x + \tan y + \tan z} = \frac{2 \sin x \sec x}{\tan x + \tan y + \tan z} \\ &= \frac{2 \tan x}{\tan x + \tan y + \tan z}\end{aligned}$$

$$\begin{aligned}\sin 2y \frac{\partial u}{\partial y} &= \sin 2y \frac{\sec^2 y}{\tan x + \tan y + \tan z} \\ &= \frac{2 \sin y \cos y \sec^2 y}{\tan x + \tan y + \tan z} = \frac{2 \sin y \sec y}{\tan x + \tan y + \tan z} \\ &= \frac{2 \tan y}{\tan x + \tan y + \tan z}\end{aligned}$$

$$\begin{aligned}\sin 2z \frac{\partial u}{\partial z} &= \sin 2z \frac{\sec^2 z}{\tan x + \tan y + \tan z} = \frac{2 \sin z \cos z \sec^2 z}{\tan x + \tan y + \tan z} \\ &= \frac{2 \sin z \sec z}{\tan x + \tan y + \tan z} = \frac{2 \tan z}{\tan x + \tan y + \tan z}\end{aligned}$$

$$\begin{aligned}\therefore \sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} &= \frac{2 \tan x}{\tan x + \tan y + \tan z} + \frac{2 \tan y}{\tan x + \tan y + \tan z} + \frac{2 \tan z}{\tan x + \tan y + \tan z} \\ &= \frac{2(\tan x + \tan y + \tan z)}{\tan x + \tan y + \tan z} \\ &= 2\end{aligned}$$

5. If $u = \frac{x^2 + y^2}{x + y}$, show that $\left(\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y}\right)^2 = 4 \left(1 - \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y}\right)$.

Solution: $u = \frac{x^2 + y^2}{x + y}$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial y} \left(\frac{x^2 + y^2}{x + y} \right) = \frac{(x + y) \frac{\partial}{\partial x} (x^2 + y^2) - (x^2 + y^2) \frac{\partial}{\partial x} (x + y)}{(x + y)^2}$$

$$= \frac{(x + y)(2x) - (x^2 + y^2)(1)}{(x + y)^2}$$

$$= \frac{2x^2 + 2xy - x^2 - y^2}{(x + y)^2} = \frac{x^2 + 2xy - y^2}{(x + y)^2}$$

$$\begin{aligned}
\frac{\partial u}{\partial y} &= \frac{\partial}{\partial y} \left(\frac{x^2 + y^2}{x + y} \right) = \frac{(x + y) \frac{\partial}{\partial y} (x^2 + y^2) - (x^2 + y^2) \frac{\partial}{\partial y} (x + y)}{(x + y)^2} \\
&= \frac{(x + y)(2y) - (x^2 + y^2)(1)}{(x + y)^2} \\
&= \frac{2xy + 2y^2 - x^2 - y^2}{(x + y)^2} \\
&= \frac{y^2 + 2xy - x^2}{(x + y)^2}
\end{aligned}$$

$$\begin{aligned}
\text{LHS} &= \left(\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \right)^2 = \left(\frac{x^2 + 2xy - y^2}{(x + y)^2} - \frac{y^2 + 2xy - x^2}{(x + y)^2} \right)^2 \\
&= \left(\frac{2x^2 - 2y^2}{(x + y)^2} \right)^2 \\
&= 4 \left(\frac{x^2 - y^2}{(x + y)^2} \right)^2 \\
&= 4 \left(\frac{(x - y)(x + y)}{(x + y)^2} \right)^2 \\
&= 4 \frac{(x - y)^2}{(x + y)^2}
\end{aligned}$$

$$\begin{aligned}
\text{RHS} &= 4 \left(1 - \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \right) \\
&= 4 \left(1 - \frac{(x^2 + 2xy - y^2)}{(x + y)^2} - \frac{(y^2 + 2xy - x^2)}{(x + y)^2} \right) \\
&= 4 \left(\frac{(x + y)^2 - (x^2 + 2xy - y^2) - (y^2 + 2xy - x^2)}{(x + y)^2} \right) \\
&= 4 \left(\frac{x^2 + 2xy + y^2 - x^2 - 2xy + y^2 - y^2 - 2xy + x^2}{(x + y)^2} \right) \\
&= 4 \left(\frac{x^2 - 2xy + y^2}{(x + y)^2} \right) = 4 \frac{(x - y)^2}{(x + y)^2}
\end{aligned}$$

$$\text{LHS} = \text{RHS}$$

6. If $u = \log(x^3 + y^3 - x^2y - y^2x)$ then show that

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right)^2 u = -\frac{4}{(x+y)^2} \quad \text{or} \quad \frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = -\frac{4}{(x+y)^2}$$

Solution: $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right)^2 u = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right) u$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\right) \dots \dots \dots (1)$$

$$\begin{aligned} u &= \log(x^3 + y^3 - x^2y - y^2x) = \log(x^3 - x^2y + y^3 - y^2x) \\ &= \log[x^2(x - y) - y^2(x - y)] \\ &= \log(x^2 - y^2)(x - y) \\ &= \log(x + y)(x - y)(x - y) = \log(x + y)(x - y)^2 \\ &= \log(x + y) + \log(x - y)^2 \\ &= \log(x + y) + 2\log(x - y) \end{aligned}$$

Differentiating u w.r.to x we get,

$$\frac{\partial u}{\partial x} = \frac{1}{x+y} + \frac{2}{x-y}$$

Differentiating u w.r.to y we get,

$$\frac{\partial u}{\partial y} = \frac{1}{x+y} - \frac{2}{x-y}$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{1}{x+y} + \frac{2}{x-y} + \frac{1}{x+y} - \frac{2}{x-y} = \frac{2}{x+y}$$

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 u = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \right) = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) \left(\frac{2}{x+y} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{2}{x+y} \right) + \frac{\partial}{\partial y} \left(\frac{2}{x+y} \right)$$

$$= -\frac{2}{(x+y)^2} - \frac{2}{(x+y)^2} = -\frac{4}{(x+y)^2}$$

7. Find the value of n if $u = r^n(3 \cos^2 \theta - 1)$ satisfies the equation

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = 0.$$

Solution: $u = r^n(3 \cos^2 \theta - 1) \dots \dots \dots (1)$

Diff.(1) w.r.to r , we get

$$\frac{\partial u}{\partial r} = nr^{n-1}(3 \cos^2 \theta - 1)$$

$$\therefore r^2 \frac{\partial u}{\partial r} = r^2 nr^{n-1}(3 \cos^2 \theta - 1) = nr^{n+1}(3 \cos^2 \theta - 1)$$

$$\Rightarrow \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = \frac{\partial}{\partial r} (nr^{n+1}(3 \cos^2 \theta - 1))$$

$$= n(n+1)r^n(3 \cos^2 \theta - 1) \dots \dots \dots (2)$$

Diff.(1) w.r.to θ , we get

$$\frac{\partial u}{\partial \theta} = r^n (-6 \cos \theta \sin \theta) = -6 r^n \sin \theta \cos \theta$$

$$\therefore \sin \theta \frac{\partial u}{\partial \theta} = -6 r^n \sin^2 \theta \cos \theta$$

$$\frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = \frac{\partial}{\partial \theta} (-6 r^n \sin^2 \theta \cos \theta)$$

$$= -6 r^n \left[\cos \theta \frac{\partial}{\partial \theta} (\sin^2 \theta) + \sin^2 \theta \frac{\partial}{\partial \theta} (\cos \theta) \right]$$

$$= -6 r^n [\cos \theta (2 \sin \theta \cos \theta) + \sin^2 \theta (-\sin \theta)]$$

$$= -6 r^n [2 \sin \theta \cos^2 \theta - \sin^3 \theta]$$

$$= -6 r^n \sin \theta [2 \cos^2 \theta - \sin^2 \theta]$$

$$\begin{aligned}
\frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) &= -6r^n \sin \theta [2\cos^2 \theta - \sin^2 \theta] \\
&= -6r^n \sin \theta [2\cos^2 \theta - (1 - \cos^2 \theta)] \\
&= -6r^n \sin \theta [3\cos^2 \theta - 1]
\end{aligned}$$

$$\begin{aligned}
\therefore \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) &= \frac{1}{\sin \theta} [-6r^n \sin \theta [3\cos^2 \theta - 1]] \\
&= -6r^n [3\cos^2 \theta - 1] \dots \dots \dots (3)
\end{aligned}$$

Now $u = r^n(3 \cos^2 \theta - 1)$ satisfies the equation

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = 0.$$

From eq.(2) and (3),

$$\therefore n(n+1)r^n(3 \cos^2 \theta - 1) - 6r^n[3 \cos^2 \theta - 1] = 0$$

$$n(n+1)r^n(3 \cos^2 \theta - 1) = 6r^n[3 \cos^2 \theta - 1]$$

$$\Rightarrow n(n+1) = 6$$

$$\Rightarrow n^2 + n - 6 = 0$$

$$\Rightarrow (n+3)(n-2) = 0 \Rightarrow n = -3, 2$$

8. Find the value of n if $\theta = t^n e^{\frac{-r^2}{4t}}$ satisfies the equation

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}.$$

Solution: $\theta = t^n e^{\frac{-r^2}{4t}}$

Taking logarithm on both sides,

$$\log \theta = \log \left(t^n e^{\frac{-r^2}{4t}} \right) = \log t^n + \log e^{\frac{-r^2}{4t}}$$

$$\log \theta = n \log t - \frac{r^2}{4t} \dots \dots \dots (1)$$

Diff.(1) w.r.to r , we get

$$\frac{1}{\theta} \frac{\partial \theta}{\partial r} = 0 - \frac{2r}{4t} = -\frac{r}{2t}$$

$$\therefore \frac{\partial \theta}{\partial r} = -\frac{r\theta}{2t}$$

$$\Rightarrow r^2 \frac{\partial \theta}{\partial r} = -\frac{r^3 \theta}{2t}$$

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial}{\partial r} \left(-\frac{r^3 \theta}{2t} \right) = -\frac{1}{2t} \left[r^3 \frac{\partial \theta}{\partial r} + \theta \frac{\partial}{\partial r} (r^3) \right]$$

$$= -\frac{1}{2t} \left[r^3 \left(-\frac{r\theta}{2t} \right) + 3r^2 \theta \right]$$

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{r^4 \theta}{4t^2} - \frac{3r^2 \theta}{2t}$$

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{r^4 \theta}{4t^2} - \frac{3r^2 \theta}{2t}$$

$$\begin{aligned} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) &= \frac{1}{r^2} \left[\frac{r^4 \theta}{4t^2} - \frac{3r^2 \theta}{2t} \right] \\ &= \frac{r^2 \theta}{4t^2} - \frac{3\theta}{2t} \dots \dots \dots (2) \end{aligned}$$

Now Diff.(1) w.r.to t ,

$$\therefore \frac{1}{\theta} \frac{\partial \theta}{\partial t} = \frac{n}{t} + \frac{r^2}{4t^2}$$
$$\Rightarrow \frac{\partial \theta}{\partial t} = \frac{n\theta}{t} + \frac{r^2\theta}{4t^2} \dots \dots \dots (3)$$

$$\theta = t^n e^{\frac{-r^2}{4t}} \text{ satisfies the equation } \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}.$$

From eq.(2) and (3),

$$\frac{r^2\theta}{4t^2} - \frac{3\theta}{2t} = \frac{n\theta}{t} + \frac{r^2\theta}{4t^2} \Rightarrow -\frac{3\theta}{2t} = \frac{n\theta}{t} \Rightarrow n = -\frac{3}{2}$$

9. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then show that

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = - \frac{9}{(x + y + z)^2}$$

Solution: $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) u$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} \right) \dots \dots \dots (1)$$

$$u = \log(x^3 + y^3 + z^3 - 3xyz)$$

$$\frac{\partial u}{\partial x} = \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz}$$

$$\frac{\partial u}{\partial y} = \frac{3y^2 - 3xz}{x^3 + y^3 + z^3 - 3xyz}$$

$$\frac{\partial u}{\partial z} = \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz}$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$$

$$= \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz} + \frac{3y^2 - 3xz}{x^3 + y^3 + z^3 - 3xyz} + \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz}$$

$$= \frac{3x^2 - 3yz + 3y^2 - 3xz + 3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz}$$

$$\begin{aligned}\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} &= \frac{3(x^2 + y^2 + z^2 - xy - yz - xz)}{x^3 + y^3 + z^3 - 3xyz} \\ &= \frac{3(x^2 + y^2 + z^2 - xy - yz - xz)}{x^3 + y^3 + z^3 - 3xyz} \frac{(x + y + z)}{(x + y + z)} \\ (x^2 + y^2 + z^2 - xy - yz - xz)(x + y + z) &= x^3 + y^3 + z^3 - 3xyz\end{aligned}$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3(x^3 + y^3 + z^3 - 3xyz)}{x^3 + y^3 + z^3 - 3xyz} \frac{1}{(x + y + z)}$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x + y + z}$$

Putting value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ in (1) we get,

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} \right)$$

$$\begin{aligned}
\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{3}{x+y+z}\right) \\
&= \frac{\partial}{\partial x} \left(\frac{3}{x+y+z}\right) + \frac{\partial}{\partial y} \left(\frac{3}{x+y+z}\right) + \frac{\partial}{\partial z} \left(\frac{3}{x+y+z}\right) \\
&= -\frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} \\
&= -\frac{9}{(x+y+z)^2}
\end{aligned}$$

Test Your Knowledge

1. If $u = x^3y - xy^3$ then find the value of $\frac{1}{\frac{\partial u}{\partial x}} + \frac{1}{\frac{\partial u}{\partial y}}$ at $(1, 2)$. **Ans:** $-\frac{13}{22}$

2. If $u = \log(x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right)$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

3. If $u = \log(e^x + e^y)$ then show that $\frac{\partial^2 u}{\partial x^2} \cdot \frac{\partial^2 u}{\partial y^2} = \left(\frac{\partial^2 u}{\partial x \partial y}\right)^2$.

4. If $z^3 - zx - y = 0$ then find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ and show that $\frac{\partial^2 z}{\partial x \partial y} = -\frac{(3z^2 + x)}{(3z^2 - x)^3}$.

5. If $u = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.

6. If $z = \tan^{-1} \left(\frac{2xy}{x^2 - y^2} \right)$ then show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.

7. If $z = \tan(y + ax) - (y - ax)^2$ then show that $\frac{\partial^2 z}{\partial x^2} = a^2 \frac{\partial^2 z}{\partial y^2}$.

8. Find the value of n if $u = Ae^{-gx} \sin(nt - gx)$ satisfies the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ where g and A are constants.

Euler's Theorem on Homogenous Functions

Homogenous function of degree n means?

A function $f(x, y)$ of two variables x and y is said to homogeneous function of degree n if

$$f(x, y) = x^n \phi\left(\frac{y}{x}\right) \quad \text{or} \quad f(x, y) = y^n \psi\left(\frac{x}{y}\right)$$

Alternately, function $f(x, y)$ of two variables x and y is said to homogeneous function of degree n if

$$f(tx, ty) = t^n f(x, y) \quad \text{where } t \text{ is a parameter.}$$

e.g.1) $f(x, y) = x^2 + y^2$ is a homogeneous function of degree 2.

$$f(tx, ty) = (tx)^2 + (ty)^2 = t^2(x^2 + y^2) = t^2 f(x, y)$$

2) $f(x, y) = \frac{x^2 y^3}{x - y}$ is a homogeneous function of degree 4.

$$f(tx, ty) = \frac{(tx)^2 (ty)^3}{tx - ty} = \frac{t^5 x^2 y^3}{t(x - y)} = t^4 f(x, y)$$

3) $f(x, y) = \log(x^2 + y^2)$ is not a homogeneous function

$$f(tx, ty) = \log((tx)^2 + (ty)^2) = \log t^2 (x^2 + y^2) \neq t^2 \log(x^2 + y^2)$$

4) $f(x, y) = \tan^{-1} \left(\frac{x}{y} \right)$ is a homogeneous function of degree 0.

$$f(tx, ty) = \tan^{-1} \left(\frac{tx}{ty} \right) = \tan^{-1} \left(\frac{x}{y} \right) = t^0 f(x, y)$$

Euler's Theorem

If u is a homogeneous function of x and y of degree n then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

Proof: u is a homogeneous function of x and y of degree n .

$$\Rightarrow u = x^n f\left(\frac{y}{x}\right)$$

$$\Rightarrow \frac{\partial u}{\partial x} = x^n f'\left(\frac{y}{x}\right) \left(-\frac{y}{x^2}\right) + nx^{n-1} f\left(\frac{y}{x}\right)$$

$$\therefore x \frac{\partial u}{\partial x} = -yx^{n-1} f'\left(\frac{y}{x}\right) + nx^n f\left(\frac{y}{x}\right) \dots \dots \dots (1)$$

$$\frac{\partial u}{\partial y} = x^n f' \left(\frac{y}{x} \right) \left(\frac{1}{x} \right)$$

$$y \frac{\partial u}{\partial y} = y x^{n-1} f' \left(\frac{y}{x} \right) \dots \dots \dots (2)$$

Adding (1) and (2), we get

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -y x^{n-1} f' \left(\frac{y}{x} \right) + n x^n f \left(\frac{y}{x} \right) + y x^{n-1} f' \left(\frac{y}{x} \right)$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n x^n f \left(\frac{y}{x} \right)$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n u \dots \dots \dots \text{Hence the proof}$$

Euler's Theorem for homogeneous function of three variables:

If u is a homogeneous function of three variables x, y and z of degree n then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$$

Deductions from Euler's Theorem:

Corollary 1:

If u is a homogeneous function of x and y of degree n then

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

1. Verify Euler's Theorem for $u = ax^2 + 2bxy + cy^2$.

Solution: $u = f(x, y) = ax^2 + 2bxy + cy^2$

Replacing x by tx and y by ty in $u = f(x, y)$,

$$f(tx, ty) = a(tx)^2 + 2btxty + c(ty)^2 = t^2(ax^2 + 2bxy + cy^2)$$

$$f(tx, ty) = t^2 f(x, y)$$

Thus, $u = ax^2 + 2bxy + cy^2$ is a homogeneous function of degree $n = 2$.

By Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u = 2(ax^2 + 2bxy + cy^2) \dots \dots \dots (1)$$

Verification: $u = ax^2 + 2bxy + cy^2$

$$\frac{\partial u}{\partial x} = 2ax + 2by$$

$$\frac{\partial u}{\partial y} = 2bx + 2cy$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = x(2ax + 2by) + y(2bx + 2cy)$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2ax^2 + 2bxy + 2bxy + 2cy^2$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2(ax^2 + 2bxy + cy^2) \dots \dots \dots (2)$$

From (1) and (2), Euler's Theorem is verified

2. Verify Euler's Theorem for $u = (\sqrt{x} + \sqrt{y})(x^n + y^n)$.

Solution: $u = f(x, y) = (\sqrt{x} + \sqrt{y})(x^n + y^n)$

Replacing x by tx and y by ty in $u = f(x, y)$,

$$f(tx, ty) = (\sqrt{tx} + \sqrt{ty})(t^n x^n + t^n y^n) = \sqrt{t} t^n (\sqrt{x} + \sqrt{y})(x^n + y^n)$$

$$f(tx, ty) = t^{n+\frac{1}{2}} f(x, y)$$

Thus, $u = (\sqrt{x} + \sqrt{y})(x^n + y^n)$ is a homogeneous function of degree $n + \frac{1}{2}$.

By Euler's Theorem,

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \left(n + \frac{1}{2} \right) u \\ \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \left(n + \frac{1}{2} \right) (\sqrt{x} + \sqrt{y})(x^n + y^n) \dots \dots \dots (1) \end{aligned}$$

Verification: $u = (\sqrt{x} + \sqrt{y})(x^n + y^n)$

$$\frac{\partial u}{\partial x} = \frac{1}{2\sqrt{x}}(x^n + y^n) + nx^{n-1}(\sqrt{x} + \sqrt{y})$$

$$\frac{\partial u}{\partial y} = \frac{1}{2\sqrt{y}}(x^n + y^n) + ny^{n-1}(\sqrt{x} + \sqrt{y})$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$$

$$= x \left[\frac{1}{2\sqrt{x}}(x^n + y^n) + nx^{n-1}(\sqrt{x} + \sqrt{y}) \right] + y \left[\frac{1}{2\sqrt{y}}(x^n + y^n) + ny^{n-1}(\sqrt{x} + \sqrt{y}) \right]$$

$$= \frac{\sqrt{x}}{2} (x^n + y^n) + nx^n (\sqrt{x} + \sqrt{y}) + \frac{\sqrt{y}}{2} (x^n + y^n) + ny^n (\sqrt{x} + \sqrt{y})$$

$$= \frac{1}{2} (\sqrt{x} + \sqrt{y}) (x^n + y^n) + n(x^n + y^n) (\sqrt{x} + \sqrt{y})$$

$$= \left(n + \frac{1}{2} \right) (\sqrt{x} + \sqrt{y}) (x^n + y^n) \dots \dots \dots (2)$$

From (1) and (2), Euler's Theorem is verified

2. Find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ if $u = 8(x^2 + y^2)(\log x - \log y)$.

Solution: $u = 8(x^2 + y^2)(\log x - \log y) = 8(x^2 + y^2) \log \left(\frac{x}{y} \right)$

Replacing x by tx and y by ty in $u = f(x, y)$,

$$\begin{aligned} f(tx, ty) &= 8[(tx)^2 + (ty)^2] \log \left(\frac{tx}{ty} \right) = t^2 \left[8(x^2 + y^2) \log \left(\frac{x}{y} \right) \right] \\ &= t^2 f(x, y) \end{aligned}$$

Therefore, u is a homogeneous function of degree $n = 2$.

By Euler's Theorem,

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= nu = 2u \\ &= 16(x^2 + y^2)(\log x - \log y). \end{aligned}$$

3. If $u = e^{\frac{x}{y}} \sin\left(\frac{x}{y}\right) + e^{\frac{y}{x}} \cos\left(\frac{y}{x}\right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.

Solution: $u = e^{\frac{x}{y}} \sin\left(\frac{x}{y}\right) + e^{\frac{y}{x}} \cos\left(\frac{y}{x}\right)$

Replacing x by tx and y by ty in $u = f(x, y)$,

$$\begin{aligned} f(tx, ty) &= e^{\frac{tx}{ty}} \sin\left(\frac{tx}{ty}\right) + e^{\frac{ty}{tx}} \cos\left(\frac{ty}{tx}\right) = e^{\frac{x}{y}} \sin\left(\frac{x}{y}\right) + e^{\frac{y}{x}} \cos\left(\frac{y}{x}\right) \\ &= t^0 f(x, y) \end{aligned}$$

Therefore, u is a homogeneous function of degree $n = 0$.

By Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu = 0u = 0$$

4. If $u = \frac{1}{x^2} + \frac{1}{y^2} + \frac{\log x - \log y}{x^2 + y^2}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + 2u = 0$.

Solution: $u = \frac{1}{x^2} + \frac{1}{y^2} + \frac{\log x - \log y}{x^2 + y^2} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{\log \left(\frac{x}{y} \right)}{x^2 + y^2}$

Replacing x by tx and y by ty in $u = f(x, y)$,

$$\begin{aligned} f(tx, ty) &= \frac{1}{(tx)^2} + \frac{1}{(ty)^2} + \frac{\log \left(\frac{tx}{ty} \right)}{(tx)^2 + (ty)^2} \\ &= \frac{1}{t^2 x^2} + \frac{1}{t^2 y^2} + \frac{\log \left(\frac{x}{y} \right)}{t^2 x^2 + t^2 y^2} \\ &= \frac{1}{t^2} \left[\frac{1}{x^2} + \frac{1}{y^2} + \frac{\log \left(\frac{x}{y} \right)}{x^2 + y^2} \right] \end{aligned}$$

$$f(tx, ty) = \frac{1}{t^2} \left[\frac{1}{x^2} + \frac{1}{y^2} + \frac{\log\left(\frac{x}{y}\right)}{x^2 + y^2} \right]$$

$$= \frac{1}{t^2} f(x, y)$$

$$f(tx, ty) = t^{-2} f(x, y)$$

Therefore, $u = f(x, y) = \frac{1}{x^2} + \frac{1}{y^2} + \frac{\log x - \log y}{x^2 + y^2}$ is a homogeneous function of degree $n = -2$.

By Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -2u \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + 2u = 0$$

5. If $u = \frac{x^3 y^3}{x^3 + y^3}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3u$
 and $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 6u$.

Solution: $u = f(x, y) = \frac{x^3 y^3}{x^3 + y^3}$

Replacing x by tx and y by ty in $u = f(x, y)$,

$$f(tx, ty) = \frac{(tx)^3 (ty)^3}{(tx)^3 + (ty)^3} = \frac{t^6 x^3 y^3}{t^3 (x^3 + y^3)} = t^3 \frac{x^3 y^3}{(x^3 + y^3)}$$

$$f(tx, ty) = t^3 f(x, y)$$

Therefore, $u = \frac{x^3 y^3}{x^3 + y^3}$ is a homogeneous function of degree $n = 3$.

By Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3u$$

$$\text{Also } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 3(3-1)u = 6u$$

6. If $u = x^2 e^{\frac{y}{x}} + y^2 \tan^{-1} \left(\frac{x}{y} \right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$
and $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2u$.

Solution: $u = f(x, y) = x^2 e^{\frac{y}{x}} + y^2 \tan^{-1} \left(\frac{x}{y} \right)$

Replacing x by tx and y by ty in $u = f(x, y)$,

$$f(tx, ty) = (tx)^2 e^{\frac{ty}{tx}} + (ty)^2 \tan^{-1} \left(\frac{tx}{ty} \right) = t^2 x^2 e^{\frac{y}{x}} + t^2 y^2 \tan^{-1} \left(\frac{x}{y} \right)$$

$$f(tx, ty) = t^2 \left[x^2 e^{\frac{y}{x}} + y^2 \tan^{-1} \left(\frac{x}{y} \right) \right] = t^2 f(x, y)$$

Thus, $u = x^2 e^{\frac{y}{x}} + y^2 \tan^{-1} \left(\frac{x}{y} \right)$ is a homogeneous function of degree $n = 2$.

By Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$$

$$\text{Also } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

$$\therefore x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2(2-1)u = 2u$$

7. If $T = \sin\left(\frac{xy}{x^2 + y^2}\right) + \sqrt{x^2 + y^2} + \frac{x^2 y}{x + y}$, find the value of $x \frac{\partial T}{\partial x} + y \frac{\partial T}{\partial y}$

Solution: Let $u = \sin\left(\frac{xy}{x^2 + y^2}\right)$, $v = \sqrt{x^2 + y^2}$, $w = \frac{x^2 y}{x + y}$

$$\therefore T = u + v + w$$

Replacing x by tx and y by ty in u , v and w ,

$$u = \sin\left(\frac{tx \, ty}{(tx)^2 + (ty)^2}\right) = \sin\left(\frac{xy}{x^2 + y^2}\right) = t^0 \sin\left(\frac{xy}{x^2 + y^2}\right)$$

$$v = \sqrt{(tx)^2 + (ty)^2} = \sqrt{t^2(x^2 + y^2)} = t\sqrt{x^2 + y^2}$$

$$w = \frac{(tx)^2 \, ty}{tx + ty} = t^2 \frac{x^2 y}{x + y}$$

Thus, u , v and w are homogeneous functions of degree 0, 1 & 2 respectively

By Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0u = 0$$

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 1v = v$$

$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = 2w$$

$$T = u + v + w \Rightarrow \frac{\partial T}{\partial x} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} + \frac{\partial w}{\partial x} \quad \text{and}$$

$$\frac{\partial T}{\partial y} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial y}$$

$$\begin{aligned}
\therefore x \frac{\partial T}{\partial x} + y \frac{\partial T}{\partial y} &= x \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} + \frac{\partial w}{\partial x} \right) + y \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial y} \right) \\
&= \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) + \left(x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} \right) + \left(x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} \right) \\
&= 0 + v + 2w \\
&= \sqrt{x^2 + y^2} + \frac{2x^2y}{x+y}
\end{aligned}$$

Test Your Knowledge

1. Verify Euler's Theorem for (i) $u = \frac{x^2 + y^2}{x + y}$ (ii) $u = x^4 y^2 \sin^{-1} \left(\frac{y}{x} \right)$

(iii) $u = 3x^2 yz + 5xy^2 z + 4z^4$ (iv) $u = \sqrt{x} + \sqrt{y} + \sqrt{z}$

2. If $u = \sin^{-1} \left[\frac{x}{y} \right] + \tan^{-1} \left[\frac{y}{x} \right]$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.

3. If $u = f \left(\frac{y}{x} \right) + \sqrt{x^2 + y^2}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sqrt{x^2 + y^2}$.

4. If $u = \frac{x}{y + z} + \frac{y}{z + x} + \frac{z}{x + y}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.

5. If $u = \frac{x^3 y^3 z^3}{x^3 + y^3 + z^3}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 6u$.

6. If $u = \log \left(\frac{\sqrt{x^2 + y^2}}{x + y} \right)$ find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$. (**Ans:** 0)

7. If $u = x^3 e^{-\frac{x}{y}}$ find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$. (**Ans:** 6)

8. If $u = \frac{x^3 + y^3}{y\sqrt{x}}$ find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ at (1,2).

9. If $u = x^3 y^2 \sin^{-1} \left(\frac{y}{x} \right)$ find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.

(**Ans:** 25u)

10. If $u = (x^2 + y^2)^{\frac{2}{3}}$ find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$. (**Ans:** $\frac{4u}{9}$)

Deductions from Euler's Theorem(cont'd)

Corollary 2: If u is not a homogeneous function of x and y but $z = f(u)$ is a homogeneous function of x and y of degree n then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \frac{f(u)}{f'(u)}$$

Corollary 3: If u is not a homogeneous function of x and y but $z = f(u)$ is a homogeneous function of x and y of degree n then

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$$

where $g(u) = n \frac{f(u)}{f'(u)}$

Deductions from Euler's Theorem

Note: If u is not a homogeneous function of x, y and z but

$w = f(u)$ is a homogeneous function of degree n in x, y and z then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = n \frac{f(u)}{f'(u)}$$

1. If $u = e^{x^2+y^2}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u \log u$.

Solution: Replacing x by tx and y by ty in u ,

$$e^{(tx)^2+(ty)^2} = e^{t^2(x^2+y^2)} \neq t^2 e^{(x^2+y^2)}$$

$\therefore u = e^{x^2+y^2}$ is not a homogeneous function of x and y .

$$\text{Now } u = e^{x^2+y^2} \Rightarrow \log u = x^2 + y^2$$

$$\text{Let } f(u) = \log u$$

Therefore, $f(u) = \log u$ is a homogeneous function of degree $n = 2$.

By Deduction of Euler's Theorem,

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{nf(u)}{f'(u)} \\ \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{2 \log u}{1/u} = 2u \log u \end{aligned}$$

2. If $u = \log \left[\frac{x^3 + y^3}{x^2 + y^2} \right]$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$ and

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -1.$$

Solution: Replacing x by tx and y by ty in u ,

$$\begin{aligned} \log \left[\frac{(tx)^3 + (ty)^3}{(tx)^2 + (ty)^2} \right] &= \log \left[\frac{t^3}{t^2} \left(\frac{x^3 + y^3}{x^2 + y^2} \right) \right] = \log \left[t \left(\frac{x^3 + y^3}{x^2 + y^2} \right) \right] \\ &\neq t \log \left[\left(\frac{x^3 + y^3}{x^2 + y^2} \right) \right] \end{aligned}$$

$\therefore u = \log \left[\frac{x^3 + y^3}{x^2 + y^2} \right]$ is not a homogeneous function of x and y .

$$\text{Now } u = \log \left[\frac{x^3 + y^3}{x^2 + y^2} \right] \Rightarrow e^u = \frac{x^3 + y^3}{x^2 + y^2}$$

$$\text{Let } f(u) = e^u$$

Therefore, $f(u) = e^u$ is a homogeneous function of degree $n = 1$.

By Deduction of Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{nf(u)}{f'(u)}$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1e^u}{e^u} = 1$$

$$\text{Also } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$$

$$\text{where } g(u) = \frac{n f(u)}{f'(u)} = \frac{1e^u}{e^u} = 1$$

$$\Rightarrow g'(u) = 0$$

$$\therefore x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 1 [0 - 1] = -1.$$

3. If $u = \sin^{-1} \left(\sqrt{x^2 + y^2} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$ and

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \tan^3 u.$$

Solution: Replacing x by tx and y by ty in u ,

$$\begin{aligned} \sin^{-1} \left(\sqrt{(tx)^2 + (ty)^2} \right) &= \sin^{-1} \left(\sqrt{t^2(x^2 + y^2)} \right) = \sin^{-1} \left(t\sqrt{x^2 + y^2} \right) \\ &\neq t \sin^{-1} \left(\sqrt{x^2 + y^2} \right) \end{aligned}$$

$\therefore u = \sin^{-1} \left(\sqrt{x^2 + y^2} \right)$ is not a homogeneous function of x and y .

$$\text{Now } u = \sin^{-1} \left(\sqrt{x^2 + y^2} \right) \Rightarrow \sin u = \sqrt{x^2 + y^2}$$

$$\text{Let } f(u) = \sin u$$

Therefore, $f(u) = \sin u$ is a homogeneous function of degree $n = 1$.

By Deduction of Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{nf(u)}{f'(u)}$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1 \sin u}{\cos u} = \tan u$$

$$\text{Also } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$$

$$\text{where } g(u) = \frac{n f(u)}{f'(u)} = \frac{1 \sin u}{\cos u} = \tan u$$

$$\Rightarrow g'(u) = \sec^2 u$$

$$\begin{aligned} \therefore x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= \tan u [\sec^2 u - 1] = \tan u [\tan^2 u] \\ &= \tan^3 u \end{aligned}$$

5. If $x = e^u \tan v$ and $y = e^u \sec v$ then find the value of $\left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}\right) \left(x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y}\right)$.

Solution: $x = e^u \tan v$ and $y = e^u \sec v$

$$\therefore y^2 - x^2 = e^{2u} \sec^2 v - e^{2u} \tan^2 v = e^{2u} (\sec^2 v - \tan^2 v) = e^{2u}$$

$$e^{2u} = y^2 - x^2$$

$$\text{Let } f(u) = e^{2u}$$

Therefore, $f(u)$ is a homogeneous function of degree $n = 2$.

By Deduction of Euler's Theorem,

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{nf(u)}{f'(u)} \\ \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{2e^{2u}}{2e^{2u}} = 1 \end{aligned}$$

Now $x = e^u \tan v$ and $y = e^u \sec v$

$$\therefore \frac{x}{y} = \frac{e^u \tan v}{e^u \sec v} = \cos v \tan v = \sin v$$

$$\Rightarrow v = \sin^{-1} \left(\frac{x}{y} \right)$$

$\Rightarrow v$ is a homogeneous function of x and y with degree $n = 0$

By Euler's Theorem,

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = nv = 0 \cdot v = 0$$

$$\therefore \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) \left(x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} \right) = 1 \times 0 = 0$$

6. If $u = \sin^{-1} \left[\frac{x+y}{\sqrt{x} + \sqrt{y}} \right]$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$ and

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{1}{4} \tan^3 u - \frac{1}{4} \tan u$$

Solution: Replacing x by tx and y by ty in u ,

$$\sin^{-1} \left[\frac{tx + ty}{\sqrt{tx} + \sqrt{ty}} \right] = \sin^{-1} \left[\frac{t}{\sqrt{t}} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right) \right] = \sin^{-1} \left[t^{1/2} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right) \right]$$

$$\neq t^{1/2} \sin^{-1} \left[\left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right) \right]$$

$\therefore u = \sin^{-1} \left[\frac{x+y}{\sqrt{x} + \sqrt{y}} \right]$ is not a homogeneous function of x and y .

$$\text{Now } u = \sin^{-1} \left[\frac{x+y}{\sqrt{x} + \sqrt{y}} \right] \Rightarrow \sin u = \frac{x+y}{\sqrt{x} + \sqrt{y}}$$

$$\text{Let } f(u) = \sin u$$

Therefore, $f(u)$ is a homogeneous function of degree $n = \frac{1}{2}$.

By Deduction of Euler's Theorem,

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{nf(u)}{f'(u)} \\ \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{1}{2} \frac{\sin u}{\cos u} = \frac{1}{2} \tan u \end{aligned}$$

$$\text{Also } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$$

$$\text{where } g(u) = \frac{nf(u)}{f'(u)} = \frac{1 \sin u}{2 \cos u} = \frac{1}{2} \tan u$$

$$\Rightarrow g'(u) = \frac{1}{2} \sec^2 u$$

$$\therefore x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{1}{2} \tan u \left[\frac{1}{2} \sec^2 u - 1 \right]$$

$$= \frac{1}{2} \tan u \left[\frac{1}{2} (1 + \tan^2 u) - 1 \right]$$

$$= \frac{1}{2} \tan u \left[\frac{1}{2} \tan^2 u - \frac{1}{2} \right]$$

$$= \frac{1}{4} \tan^3 u - \frac{1}{4} \tan u$$

7. If $u = \operatorname{cosec}^{-1} \left[\sqrt{\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}} \right]$, prove that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144} (13 + \tan^2 u).$$

Solution: Replacing x by tx and y by ty in u ,

$$\begin{aligned} \operatorname{cosec}^{-1} \left[\sqrt{\frac{(tx)^{1/2} + (ty)^{1/2}}{(tx)^{1/3} + (ty)^{1/3}}} \right] &= \operatorname{cosec}^{-1} \left[\sqrt{\frac{t^{1/2}}{t^{1/3}} \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)} \right] = \operatorname{cosec}^{-1} \left[\sqrt{t^{\frac{1}{6}} \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)} \right] \\ &= \operatorname{cosec}^{-1} \left[\frac{1}{t^{1/2}} \sqrt{\left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)} \right] \neq \frac{1}{t^{1/2}} \operatorname{cosec}^{-1} \left[\sqrt{\left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)} \right] \end{aligned}$$

$\therefore u = \operatorname{cosec}^{-1} \left[\sqrt{\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}} \right]$ is not a homogeneous function of x and y .

$$\text{Now } u = \operatorname{cosec}^{-1} \left[\sqrt{\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}} \right] \Rightarrow \operatorname{cosec} u = \sqrt{\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}}$$

Let $f(u) = \operatorname{cosec} u$

Therefore, $f(u)$ is a homogeneous function of degree $n = \frac{1}{12}$.

By Deduction of Euler's Theorem,

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1] \quad \text{where } g(u) = \frac{n f(u)}{f'(u)}$$

$$g(u) = \frac{n f(u)}{f'(u)} = \frac{1}{12} \frac{\operatorname{cosec} u}{(-\operatorname{cosec} u \cot u)} = -\frac{1}{12} \tan u$$

$$\Rightarrow g'(u) = -\frac{1}{12} \sec^2 u$$

$$\Rightarrow x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{1}{12} \tan u \left[-\frac{1}{12} \sec^2 u - 1 \right]$$

$$= \frac{1}{12} \tan u \left[\frac{1}{12} (1 + \tan^2 u) + 1 \right]$$

$$= \frac{1}{12} \tan u \left[\frac{1}{12} \tan^2 u + \frac{13}{12} \right]$$

$$= \frac{\tan u}{144} [\tan^2 u + 13]$$

8. If $u = \sec^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$.

Also find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$.

Solution: Replacing x by tx and y by ty in u ,

$$\sec^{-1} \left(\frac{(tx)^3 + (ty)^3}{tx + ty} \right) \\ = \sec^{-1} \left(\frac{t^3}{t} \frac{x^3 + y^3}{x + y} \right) = \sec^{-1} \left(t^2 \frac{x^3 + y^3}{x + y} \right) \neq t^2 \sec^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$$

$\therefore u = \sec^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$ is not a homogeneous function of x and y .

$$u = \sec^{-1} \left(\frac{x^3 + y^3}{x + y} \right) \Rightarrow \sec u = \frac{x^3 + y^3}{x + y}$$

Let $f(u) = \sec u$

Therefore, $f(u) = \sec u$ is a homogeneous function of degree $n = 2$.

By Deduction of Euler's Theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{n f(u)}{f'(u)}$$

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{2 \sec u}{\sec u \tan u} = \frac{2}{\tan u} \\ &= 2 \cot u \end{aligned}$$

$$\text{Also } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$$

$$\text{where } g(u) = \frac{nf(u)}{f'(u)} = \frac{2 \sec u}{\sec u \tan u} = \frac{2}{\tan u} = 2 \cot u$$

$$\Rightarrow g'(u) = -2 \operatorname{cosec}^2 u$$

$$\therefore x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \cot u [-2 \operatorname{cosec}^2 u - 1]$$

$$= 2 \cot u [-2 (1 + \cot^2 u) - 1]$$

$$= 2 \cot u [-2 - 2 \cot^2 u - 1]$$

$$= 2 \cot u [-3 - 2 \cot^2 u]$$

$$= -6 \cot u - 4 \cot^3 u$$

Test Your Knowledge

1. If $u = \cos^{-1} \left[\frac{x+y}{\sqrt{x} + \sqrt{y}} \right]$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \cot u$.
2. If $u = \sin^{-1} \left[\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}} \right]$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = -3 \tan u$
3. If $u = \sin^{-1} \left[\frac{x+y}{\sqrt{x} + \sqrt{y}} \right]$ then show that
$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{\sin u \cos 2u}{\cos^3 u}$$
4. If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$ then show that
$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = (1 - 4\sin^2 u) \sin 2u$$

5. If $u = \sin^{-1} \left[\sqrt{\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}} \right]$ then prove that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144} (13 + \tan^2 u).$$

6. If $u = \sin^{-1}(x^3 + y^3)^{2/5}$ then show that

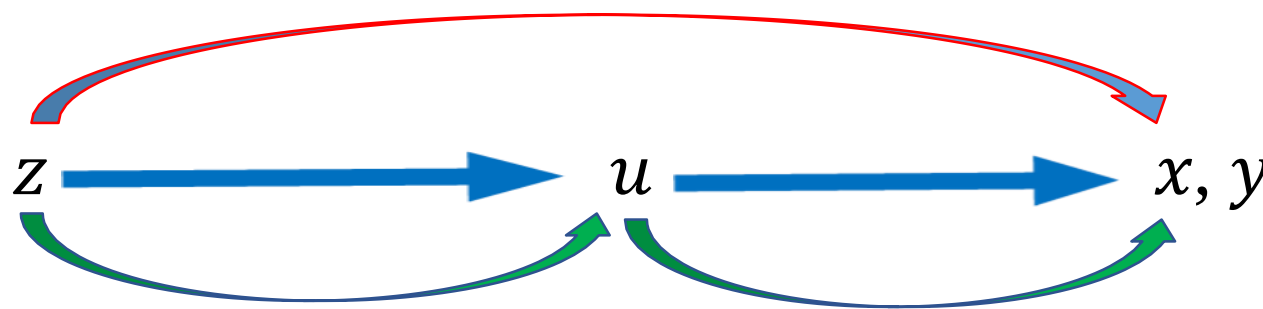
$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{6}{5} \tan u \left(\frac{6}{5} \sec^2 u - 1 \right)$$

7. If $u = \tan^{-1} \left(\frac{y^2}{x} \right)$ then show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\sin^2 u \sin 2u$

8. If $u = \sin^{-1}(xyz)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = -3 \tan u$

Partial Derivative of Composite Function

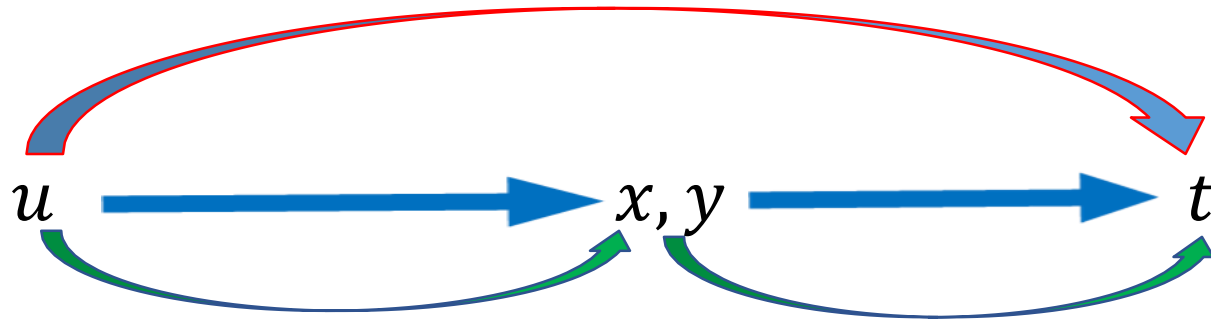
If $z = f(u)$ is a function of u and u is again a function of two variables x and y , i. e., $u = \phi(x, y)$ then z is called the composite function of x and y .



$$\frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{df}{du} \cdot \frac{\partial u}{\partial x} = f'(u) \frac{\partial u}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{df}{du} \cdot \frac{\partial u}{\partial y} = f'(u) \frac{\partial u}{\partial y}$$

If $u = f(x, y)$ and $x = \phi(t)$, $y = \psi(t)$ then u is called the **composite function** of a single variable t .



$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$\frac{du}{dt}$ is called as **total derivative** of u .

If $u = f(x, y, z)$ and $x = \phi(t)$, $y = \psi(t)$, $z = \xi(t)$ then total derivative of u is

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

1. If $u = y^2 - 4ax$, $x = at^2$, $y = 2at$ then find $\frac{du}{dt}$.

Solution: $u = y^2 - 4ax$, $x = at^2$, $y = 2at$

Here u is a function of x and y . Also x and y are functions of t .

Therefore u is the composite function of a single variable t .

$$\begin{aligned}\therefore \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \\ &= (-4a)(2at) + (2y)(2a) \\ &= -8a^2t + 2(2at)(2a) \\ &= -8a^2t + 8a^2t \\ &= 0\end{aligned}$$

2. If $u = \sin\left(\frac{x}{y}\right)$, $x = e^t$, $y = t^2$ then find $\frac{du}{dt}$.

Solution: Here u is a function of x and y . Also x and y are functions of t . Therefore u is the composite function of a single variable t .

$$\therefore \frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$= \frac{1}{y} \cos\left(\frac{x}{y}\right) \cdot (e^t) + \left(-\frac{x}{y^2}\right) \cos\left(\frac{x}{y}\right) \cdot (2t)$$

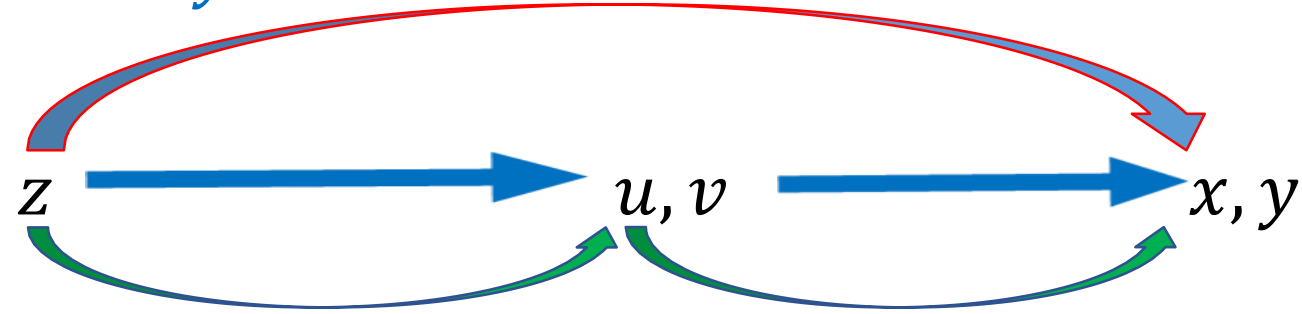
$$= \frac{1}{t^2} \cos\left(\frac{e^t}{t^2}\right) \cdot (e^t) + \left(-\frac{e^t}{(t^2)^2}\right) \cos\left(\frac{e^t}{t^2}\right) \cdot (2t)$$

$$\frac{du}{dt} = \frac{e^t}{t^2} \cos\left(\frac{e^t}{t^2}\right) + \left(-\frac{2e^t}{t^3}\right) \cos\left(\frac{e^t}{t^2}\right)$$

$$= \frac{e^t}{t^2} \cos\left(\frac{e^t}{t^2}\right) \cdot \left(1 - \frac{2}{t}\right)$$

Composite Function of Two Variables

If $z = f(u, v)$ and $u = \phi(x, y)$, $v = \psi(x, y)$ then z is called the **composite function of two variables x and y** .



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

1. If $z = f(u, v)$, $u = \log(x^2 + y^2)$, $v = \frac{y}{x}$ then show that

$$x \frac{\partial z}{\partial y} - y \frac{\partial z}{\partial x} = (1 + v^2) \frac{\partial z}{\partial v}.$$

Solution: $z = f(u, v)$, $u = \log(x^2 + y^2)$, $v = \frac{y}{x}$

Here z is a function of u and v . Also u and v both are functions of x and y .

Therefore z is a composite function of two variables x and y .

$$\therefore \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \quad \text{and}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \left(\frac{1}{x^2 + y^2} \cdot 2x \right) + \frac{\partial z}{\partial v} \cdot \left(-\frac{y}{x^2} \right) = \frac{2x}{x^2 + y^2} \frac{\partial z}{\partial u} - \frac{y}{x^2} \frac{\partial z}{\partial v}$$

$$\therefore y \frac{\partial z}{\partial x} = \frac{2xy}{x^2 + y^2} \frac{\partial z}{\partial u} - \frac{y^2}{x^2} \frac{\partial z}{\partial v} \dots \dots \dots (1)$$

$$\text{Now } \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$= \frac{\partial z}{\partial u} \cdot \left(\frac{1}{x^2 + y^2} \cdot 2y \right) + \frac{\partial z}{\partial v} \cdot \left(\frac{1}{x} \right) = \frac{2y}{x^2 + y^2} \frac{\partial z}{\partial u} + \frac{1}{x} \frac{\partial z}{\partial v}$$

$$\frac{\partial z}{\partial y} = \frac{2y}{x^2 + y^2} \frac{\partial z}{\partial u} + \frac{1}{x} \frac{\partial z}{\partial v}$$

$$\therefore x \frac{\partial z}{\partial y} = \frac{2xy}{x^2 + y^2} \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \dots \dots \dots (2)$$

Subtracting Eq.(1) from (2),

$$LHS = x \frac{\partial z}{\partial y} - y \frac{\partial z}{\partial x} = \left[\frac{2xy}{x^2 + y^2} \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right] - \left[\frac{2xy}{x^2 + y^2} \frac{\partial z}{\partial u} - \frac{y^2}{x^2} \frac{\partial z}{\partial v} \right]$$

$$= \left(1 + \frac{y^2}{x^2} \right) \frac{\partial z}{\partial v}$$

$$= (1 + v^2) \frac{\partial z}{\partial v} = RHS$$

2. If $z = f(u, v)$, $u = x^2 + y^2$, $v = 2xy$ then show that

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = 2\sqrt{u^2 - v^2} \frac{\partial z}{\partial u}.$$

Solution: $z = f(u, v)$, $u = x^2 + y^2$, $v = 2xy$

Here z is a function of u and v . Also u and v both are functions of x and y .

Therefore z is a composite function of two variables x and y .

$$\therefore \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \quad \text{and}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot (2x) + \frac{\partial z}{\partial v} \cdot (2y)$$

$$\therefore x \frac{\partial z}{\partial x} = 2x^2 \frac{\partial z}{\partial u} + 2xy \frac{\partial z}{\partial v} \dots \dots \dots (1)$$

$$\text{Now } \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$= \frac{\partial z}{\partial u} \cdot (2y) + \frac{\partial z}{\partial v} \cdot (2x)$$

$$\therefore y \frac{\partial z}{\partial y} = 2y^2 \frac{\partial z}{\partial u} + 2xy \frac{\partial z}{\partial v} \dots \dots \dots (2)$$

Subtracting Eq.(2) from (1),

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = \left[2x^2 \frac{\partial z}{\partial u} + 2xy \frac{\partial z}{\partial v} \right] - \left[2y^2 \frac{\partial z}{\partial u} + 2xy \frac{\partial z}{\partial v} \right]$$

$$= 2(x^2 - y^2) \frac{\partial z}{\partial u}$$

$$= 2\sqrt{(x^2 - y^2)^2} \frac{\partial z}{\partial u}$$

$$= 2\sqrt{(x^2 + y^2)^2 - 4x^2y^2} \frac{\partial z}{\partial u}$$

$$= 2\sqrt{u^2 - v^2} \frac{\partial z}{\partial u}$$

3. If $z = f(u, v)$, $u = x \cos \theta - y \sin \theta$, $v = x \sin \theta + y \cos \theta$ then show that

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v}.$$

Solution: $z = f(u, v)$, $u = x \cos \theta - y \sin \theta$, $v = x \sin \theta + y \cos \theta$.

Here z is a function of u and v . Also u and v both are functions of x and y .

Therefore z is a composite function of two variables x and y .

$$\therefore \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \quad \text{and}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} (\cos \theta) + \frac{\partial z}{\partial v} (\sin \theta)$$

$$\therefore x \frac{\partial z}{\partial x} = x \cos \theta \frac{\partial z}{\partial u} + x \sin \theta \frac{\partial z}{\partial v} \dots \dots \dots (1)$$

$$\text{Now } \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$= \frac{\partial z}{\partial u} (-\sin \theta) + \frac{\partial z}{\partial v} (\cos \theta)$$

$$\therefore y \frac{\partial z}{\partial y} = -y \sin \theta \frac{\partial z}{\partial u} + y \cos \theta \frac{\partial z}{\partial v} \dots \dots \dots (2)$$

Adding Eqs.(1) and (2),

$$\begin{aligned}x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= \left[x \cos \theta \frac{\partial z}{\partial u} + x \sin \theta \frac{\partial z}{\partial v} \right] - \left[-y \sin \theta \frac{\partial z}{\partial u} + y \cos \theta \frac{\partial z}{\partial v} \right] \\&= (x \cos \theta + y \sin \theta) \frac{\partial z}{\partial u} + (x \sin \theta - y \cos \theta) \frac{\partial z}{\partial v} \\&= u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v}\end{aligned}$$

4. If $z = f(x, y)$, $x = e^u + e^{-v}$, $y = e^{-u} - e^v$ then show that

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}.$$

Solution: $z = f(x, y)$, $x = e^u + e^{-v}$, $y = e^{-u} - e^v$

Here z is a function of x and y . Also x and y both are functions of u and v .

Therefore z is a composite function of two variables u and v .

$$\therefore \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \quad \text{and}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} (e^u) + \frac{\partial z}{\partial y} (-e^{-u}) \dots \dots \dots (1)$$

Now

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} (-e^{-v}) + \frac{\partial z}{\partial y} (-e^v) \dots \dots \dots (2)$$

Subtracting Eq.(2) from (1),

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = \left[\frac{\partial z}{\partial x} (e^u) + \frac{\partial z}{\partial y} (-e^{-u}) \right] - \left[\frac{\partial z}{\partial x} (-e^{-v}) + \frac{\partial z}{\partial y} (-e^v) \right]$$

$$\begin{aligned}
\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} &= \left[\frac{\partial z}{\partial x}(e^u) + \frac{\partial z}{\partial y}(-e^{-u}) \right] - \left[\frac{\partial z}{\partial x}(-e^{-v}) + \frac{\partial z}{\partial y}(-e^v) \right] \\
&= (e^u + e^{-v}) \frac{\partial z}{\partial x} - (e^{-u} - e^v) \frac{\partial z}{\partial x} \\
&= x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial x}
\end{aligned}$$

5. If $z = f(x, y)$, $x^2 = au + bv$, $y^2 = au - bv$ then show that

$$u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v} = \frac{1}{2} \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right).$$

Solution: $z = f(x, y)$, $x^2 = au + bv \Rightarrow x = \sqrt{au + bv}$,

$$y^2 = au - bv \Rightarrow y = \sqrt{au - bv}$$

Here z is a function of x and y . Also x and y both are functions of u and v .

Therefore z is a composite function of two variables u and v .

$$\therefore \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \quad \text{and}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \left(\frac{a}{2\sqrt{au + bv}} \right) + \frac{\partial z}{\partial y} \left(\frac{a}{2\sqrt{au - bv}} \right)$$

$$\frac{\partial z}{\partial u} = \frac{a}{2x} \frac{\partial z}{\partial x} + \frac{a}{2y} \frac{\partial z}{\partial y}$$

$$\therefore u \frac{\partial z}{\partial u} = \frac{au}{2x} \frac{\partial z}{\partial x} + \frac{au}{2y} \frac{\partial z}{\partial y} \dots \dots \dots (1)$$

$$\text{Now } \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \left(\frac{b}{2\sqrt{au + bv}} \right) + \frac{\partial z}{\partial y} \left(\frac{-b}{2\sqrt{au - bv}} \right)$$

$$\frac{\partial z}{\partial v} = \frac{b}{2x} \frac{\partial z}{\partial x} - \frac{b}{2y} \frac{\partial z}{\partial y}$$

$$\therefore v \frac{\partial z}{\partial v} = \frac{bv}{2x} \frac{\partial z}{\partial x} - \frac{bv}{2y} \frac{\partial z}{\partial y} \dots \dots \dots (2)$$

Adding Eqs.(1) and (2),

$$\begin{aligned}u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v} &= \left[\frac{au}{2x} \frac{\partial z}{\partial x} + \frac{av}{2y} \frac{\partial z}{\partial y} \right] + \left[\frac{bv}{2x} \frac{\partial z}{\partial x} - \frac{bv}{2y} \frac{\partial z}{\partial y} \right] \\&= \left(\frac{au + bv}{2x} \right) \frac{\partial z}{\partial x} + \left(\frac{av - bv}{2y} \right) \frac{\partial z}{\partial y} \\&= \left(\frac{x^2}{2x} \right) \frac{\partial z}{\partial x} + \left(\frac{y^2}{2y} \right) \frac{\partial z}{\partial y} \\&= \frac{1}{2} \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right)\end{aligned}$$

6. If $u = f(x - y, y - z, z - x)$, then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

Solution: Let $x - y = l$, $y - z = m$, $z - x = n$

$$u = f(x - y, y - z, z - x) = f(l, m, n)$$

Here u is a function of l, m and n . Also l, m and n are functions of x, y and z .

Therefore u is a composite function of three variables x, y and z .

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial x} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial x} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial y} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial y} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial y}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial z} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial z} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial z}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial x} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial x} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial l} (1) + \frac{\partial u}{\partial m} (0) + \frac{\partial u}{\partial n} (-1)$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial l} - \frac{\partial u}{\partial n} \dots \dots \dots (1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial y} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial y} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial l} (-1) + \frac{\partial u}{\partial m} (1) + \frac{\partial u}{\partial n} (0)$$

$$\frac{\partial u}{\partial y} = -\frac{\partial u}{\partial l} + \frac{\partial u}{\partial m} \dots \dots \dots (2)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial z} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial z} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial z}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial l} (0) + \frac{\partial u}{\partial m} (-1) + \frac{\partial u}{\partial n} (1)$$

$$\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial m} + \frac{\partial u}{\partial n} \dots \dots \dots (3)$$

Adding Eqs.(1), (2) and (3),

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \left(\frac{\partial u}{\partial l} - \frac{\partial u}{\partial n} \right) + \left(-\frac{\partial u}{\partial l} + \frac{\partial u}{\partial m} \right) + \left(-\frac{\partial u}{\partial m} + \frac{\partial u}{\partial n} \right)$$

$$\Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

7. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.

Solution: Let $\frac{x}{y} = l, \quad \frac{y}{z} = m, \quad \frac{z}{x} = n$

$$u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right) = f(l, m, n)$$

Here u is a function of l, m and n . Also l, m and n are functions of x, y and z .

Therefore u is a composite function of three variables x, y and z .

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial x} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial x} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial y} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial y} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial y}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial z} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial z} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial z}$$

$$\text{Now } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial l} \left(\frac{1}{y} \right) + \frac{\partial u}{\partial m} (0) + \frac{\partial u}{\partial n} \left(-\frac{z}{x^2} \right)$$

$$\therefore x \frac{\partial u}{\partial x} = \frac{x}{y} \frac{\partial u}{\partial l} - \frac{z}{x} \frac{\partial u}{\partial n} \dots \dots \dots (1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial l} \left(-\frac{x}{y^2} \right) + \frac{\partial u}{\partial m} \left(\frac{1}{z} \right) + \frac{\partial u}{\partial n} (0)$$

$$\therefore y \frac{\partial u}{\partial y} = -\frac{x}{y} \frac{\partial u}{\partial l} + \frac{y}{z} \frac{\partial u}{\partial m} \dots \dots \dots (2)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial l} (0) + \frac{\partial u}{\partial m} \left(-\frac{y}{z^2} \right) + \frac{\partial u}{\partial n} \left(\frac{1}{x} \right)$$

$$\therefore z \frac{\partial u}{\partial z} = -\frac{y}{z} \frac{\partial u}{\partial m} + \frac{z}{x} \frac{\partial u}{\partial n} \dots \dots \dots (2)$$

Adding Eqs.(1), (2) and (3),

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$$

$$= \left(\frac{x \partial u}{y \partial l} - \frac{z \partial u}{x \partial n} \right) + \left(-\frac{x \partial u}{y \partial l} + \frac{y \partial u}{z \partial m} \right) + \left(-\frac{y \partial u}{z \partial m} + \frac{z \partial u}{x \partial n} \right)$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$$

Implicit Functions

A function $f(x, y) = c$ is called an implicit function if y is a function of x but y can not be expressed in terms of x .

e. g. $x^3 + y^3 = 3axy$, $y \sin x = x \cos y$, $x^y + y^x = c$ are implicit functions.

If $f(x, y) = c$ is an implicit function then

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \quad \text{and}$$

$$\frac{d^2y}{dx^2} = \frac{1}{q^3} \begin{vmatrix} r & s & p \\ s & t & q \\ p & q & 0 \end{vmatrix} = -\frac{1}{q^3} [q^2r - 2pq s + p^2t]$$

$$\text{where } p = \frac{\partial f}{\partial x}, \quad q = \frac{\partial f}{\partial y}, \quad r = \frac{\partial^2 f}{\partial x^2}, \quad s = \frac{\partial^2 f}{\partial x \partial y} \quad \text{and} \quad t = \frac{\partial^2 f}{\partial y^2}$$

Q. Find $\frac{dy}{dx}$ if (i) $x^3 + y^3 = 3axy$, (ii) $y \sin x = x \cos y$, (iii) $x^y + y^x = c$

Solution: (i) Let $f(x, y) = x^3 + y^3 - 3axy$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{(3x^2 - 3ay)}{(3y^2 - 3ax)} = \frac{ay - x^2}{y^2 - ax}$$

(ii) Let $f(x, y) = y \sin x - x \cos y$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{(y \cos x - \cos y)}{(\sin x + x \sin y)} = \frac{\cos y - y \cos x}{\sin x + x \sin y}$$

(ii) Let $f(x, y) = x^y + y^x - c$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{(yx^{y-1} + y^x \log y)}{(x^y \log x + x^{y-1})}$$

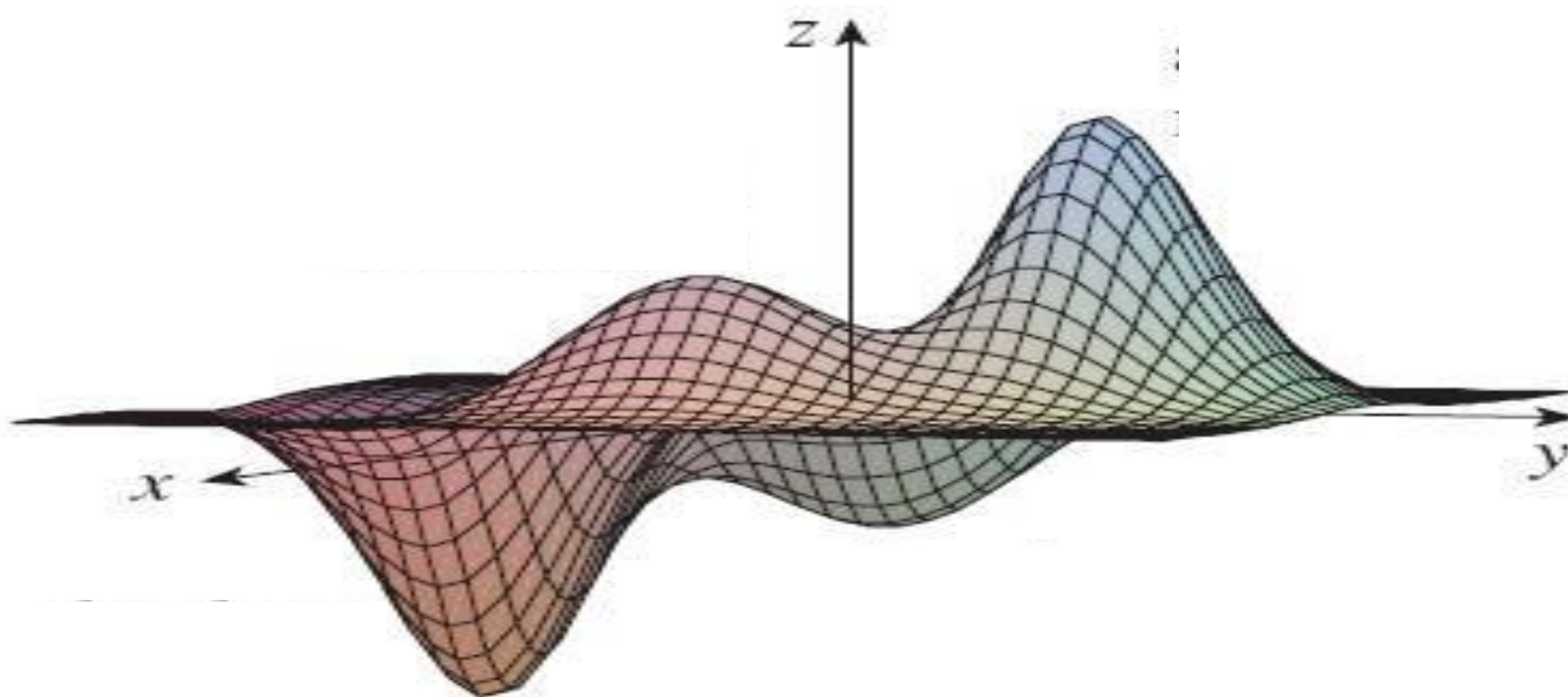
Applications of Partial Derivatives

Contents

1. Maxima and Minima of function of two variables
2. Lagrange's Method of undetermined multiplier
3. Errors & Approximations
4. Jacobian

1. Maxima and Minima of function of two variables

Look at the hills and valleys in the graph of f shown here.

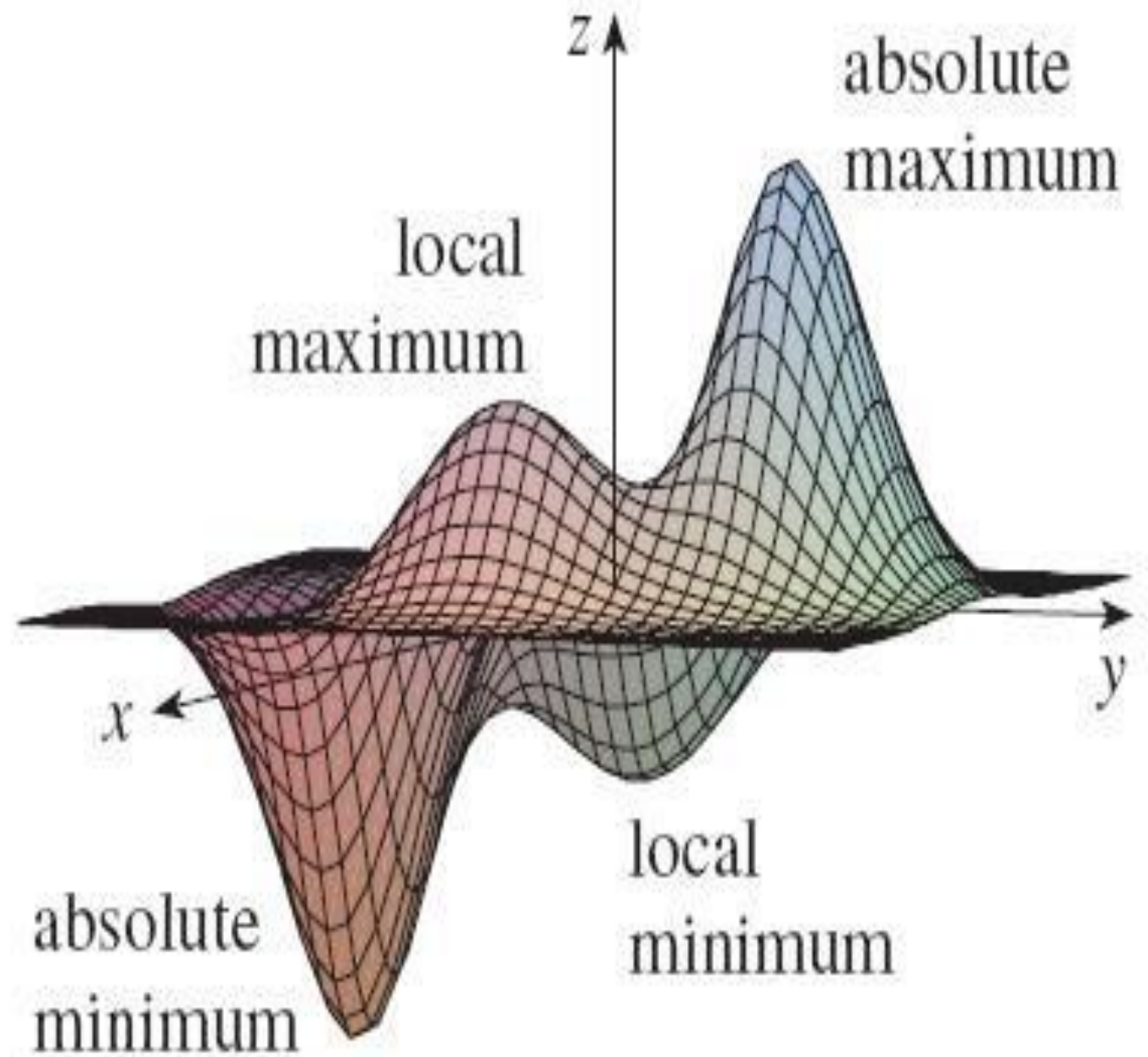


There are two points where f has a local maximum.

The larger of these two values is the **absolute maximum**.

Similarly there are two points where f has a local minimum.

The smaller of these two values is the **absolute minimum**.



Let f be a function defined on a region R containing the point (a, b) . Then

1. f has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ for all points (x, y) that are very near to (a, b) .

The number $f(a, b)$ is called a **local maximum value**.

2. If $f(x, y) \leq f(a, b)$ for all points (x, y) in the domain R of the function f then number $f(a, b)$ is called a **absolute maximum value**.

3. f has a **local minimum** at (a, b) if $f(x, y) \geq f(a, b)$ for all points (x, y) that are very near to (a, b) .

The number $f(a, b)$ is called a **local minimum value**.

4. If $f(x, y) \geq f(a, b)$ for all points (x, y) in the domain R of the function f then number $f(a, b)$ is called a **absolute minimum value**.

Working rule for finding Extreme (Maximum/Minimum) values

Let $f(x, y)$ be a given function of x & y .

1. Find partial derivatives $p = \frac{\partial f}{\partial x}$, $q = \frac{\partial f}{\partial y}$, $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$.

2. Find stationary points by solving equations $p = \frac{\partial f}{\partial x} = 0$ & $q = \frac{\partial f}{\partial y} = 0$

simultaneously. Suppose (a, b) be the stationary point.

3. Find the values of $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$ at point (a, b) .

4. Find $rt - s^2$ at point (a, b) . Then

a) $rt - s^2 > 0$ and $r < 0 \Rightarrow f(x, y)$ has **maximum** value at the point (a, b) .

b) $rt - s^2 > 0$ and $r > 0 \Rightarrow f(x, y)$ has **minimum** value at the point (a, b) .

c) $rt - s^2 < 0 \Rightarrow f(x, y)$ has **neither a maximum nor a minimum** value at the point (a, b) . Such a point is called as a **saddle point**.

d) $rt - s^2 = 0$ then test gives no information. Further investigation is needed.

Note:

1. The point (a, b) at which $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$ is called as **stationary point**.

2. Maximum/Minimum values occur only at stationary points.

However stationary points need not be maxima/minima.

Given function $f(x, y)$

Find $p = \frac{\partial f}{\partial x}$, $q = \frac{\partial f}{\partial y}$, $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$

Find stationary points by solving equations $p = \frac{\partial f}{\partial x} = 0$ & $q = \frac{\partial f}{\partial y} = 0$ simultaneously

Calculate the value of $rt - s^2$ at stationary points

$rt - s^2 > 0$

$r > 0$

Minima

$r < 0$

Maxima

$rt - s^2 < 0$

Neither maxima nor minima. Such a point is a saddle point

$rt - s^2 = 0$

Test fails. Further investigation is required

Example 1. Find extreme values of the function $f(x, y) = x^2 + y^2$.

Solution : $f(x, y) = x^2 + y^2$

$$p = \frac{\partial f}{\partial x} = 2x,$$

$$r = \frac{\partial^2 f}{\partial x^2} = 2, \quad s = \frac{\partial^2 f}{\partial x \partial y} = 0,$$

$$q = \frac{\partial f}{\partial y} = 2y$$

$$t = \frac{\partial^2 f}{\partial y^2} = 2$$

$$p = 0 \Rightarrow 2x = 0 \Rightarrow x = 0$$

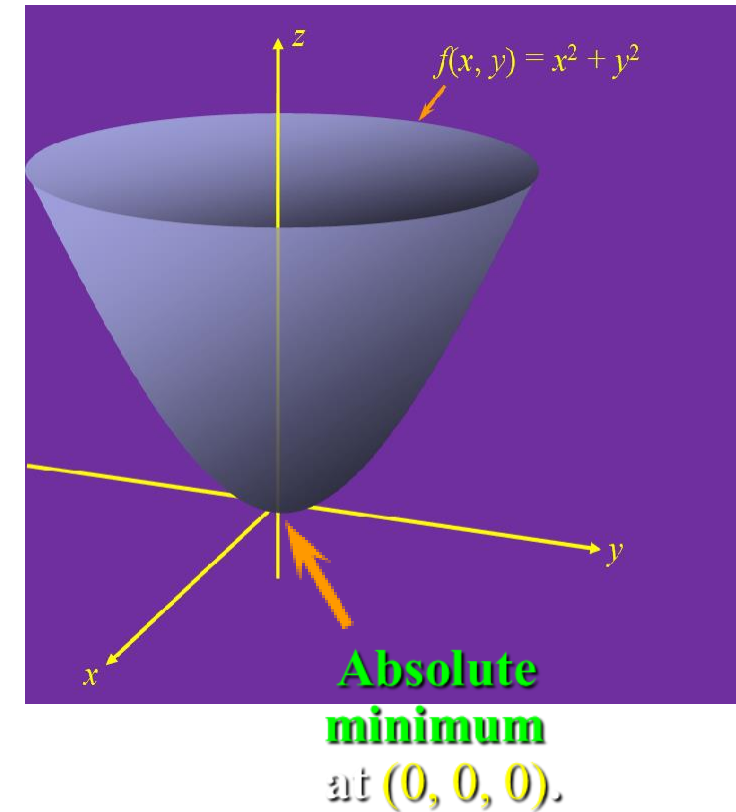
$$q = 0 \Rightarrow 2y = 0 \Rightarrow y = 0$$

Stationary point is $(0, 0)$.

At point $(0, 0)$, $rt - s^2 = (2)(2) - (0)^2 = 4 > 0$ and $r = 2 > 0$.

$f(x, y)$ has minimum value at the point $(0, 0)$.

$$f_{min} = f(0, 0) = 0$$



Example 2. Find extreme values of the function $f(x, y) = 1 - x^2 - y^2$.

Solution : $f(x, y) = 1 - x^2 - y^2$

$$p = \frac{\partial f}{\partial x} = -2x,$$

$$q = \frac{\partial f}{\partial y} = -2y$$

$$r = \frac{\partial^2 f}{\partial x^2} = -2, \quad s = \frac{\partial^2 f}{\partial x \partial y} = 0, \quad t = \frac{\partial^2 f}{\partial y^2} = -2$$

$$p = 0 \implies -2x = 0 \implies x = 0$$

$$q = 0 \implies -2y = 0 \implies y = 0$$

Stationary point is $(0, 0)$.

At point $(0, 0)$, $rt - s^2 = (-2)(-2) - (0)^2 = 4 > 0$ and $r = -2 < 0$.

$f(x, y)$ has maximum value at the point $(0, 0)$.

$$f_{\max} = f(0, 0) = 1$$

Example 3. Find extreme values of the function $f(x, y) = xy$.

Solution : $f(x, y) = xy$

$$p = \frac{\partial f}{\partial x} = y,$$

$$q = \frac{\partial f}{\partial y} = x$$

$$r = \frac{\partial^2 f}{\partial x^2} = 0,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 1,$$

$$t = \frac{\partial^2 f}{\partial y^2} = 0$$

$$p = 0 \implies y = 0 \quad \& \quad q = 0 \implies x = 0$$

Stationary point is $(0, 0)$.

At point $(0, 0)$, $rt - s^2 = (0)(0) - 1^2 = -1 < 0$.

$f(x, y)$ has neither maxima nor minima at point $(0, 0)$.

So, $(0, 0)$ is a saddle point of $f(x, y)$.

Example 4. Find extreme values of the function $f(x, y) = 1 - x^2y^2$.

Solution : $f(x, y) = 1 - x^2y^2$

$$p = \frac{\partial f}{\partial x} = -2xy^2,$$

$$q = \frac{\partial f}{\partial y} = -2x^2y$$

$$r = \frac{\partial^2 f}{\partial x^2} = -2y^2,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -4xy,$$

$$t = \frac{\partial^2 f}{\partial y^2} = -2x^2$$

$$p = 0 \Rightarrow -2xy^2 = 0 \Rightarrow x = 0, y = 0$$

$$q = 0 \Rightarrow -2x^2y = 0 \Rightarrow x = 0, y = 0$$

Stationary point is $(0, 0)$.

$$rt - s^2 = (-2y^2)(-2x^2) - (-4xy)^2 = 4x^2y^2 - 16x^2y^2 = -12x^2y^2 = 0$$

$$\text{At point } (0, 0), \quad rt - s^2 = 0$$

Test failed. Further investigation is needed.

Example 5. Find extreme values of $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$

Solution : $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$

$$p = \frac{\partial f}{\partial x} = y - 2x - 2,$$

$$q = \frac{\partial f}{\partial y} = x - 2y - 2$$

$$r = \frac{\partial^2 f}{\partial x^2} = -2,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 1,$$

$$t = \frac{\partial^2 f}{\partial y^2} = -2$$

$$p = 0 \Rightarrow y - 2x - 2 = 0 \dots\dots\dots (1)$$

$$q = 0 \Rightarrow x - 2y - 2 = 0 \dots\dots\dots (2)$$

Solving (1) and (2) simultaneously we get, $x = -2$ and $y = -2$

Therefore stationary point is ()

$$\text{Thus } rt - s^2 = (-2)(-2) - (1)^2 = 4 - 1 = 3$$

At point $(-2, -2)$, $rt - s^2 = 3 > 0$ and $r = -2 < 0$.

$\Rightarrow f(x, y)$ has maximum value at the point $(-2, -2)$.

$$f_{max} = f(-2, -2)$$

$$= (-2)(-2) - (-2)^2 - (-2)^2 - 2(-2) - 2(-2) + 4$$

$$= 4 - 4 - 4 + 4 + 4 + 4$$

$$= 8$$

Example 6. Find extreme values of the function $f(x, y) = x^2 + y^2 - 4y + 9$

Solution : $f(x, y) = x^2 + y^2 - 4y + 9$

$$p = \frac{\partial f}{\partial x} = 2x,$$

$$q = \frac{\partial f}{\partial y} = 2y - 4$$

$$r = \frac{\partial^2 f}{\partial x^2} = 2,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0,$$

$$t = \frac{\partial^2 f}{\partial y^2} = 2$$

$$p = 0 \Rightarrow 2x = 0 \Rightarrow x = 0 \dots \dots \dots (1)$$

$$q = 0 \Rightarrow 2y - 4 = 0 \Rightarrow y = 2 \dots \dots \dots (2)$$

From (1) and (2), stationary point is $(0, 2)$.

$$\text{Thus } rt - s^2 = (2)(2) - (0)^2 = 4$$

At point $(0, 2)$, $rt - s^2 = 4 > 0$ and $r = 2 > 0$.

$\Rightarrow f(x, y)$ has minimum value at the point $(0, 2)$.

$$f_{\min} = f(0, 2) = 0 + (2)^2 - 4(2) + 9 = 5$$

Example 7. Find extreme values of $f(x, y) = x^2 + 4y^3 - 12y^2 - 36y + 2$

Solution : $f(x, y) = x^2 + 4y^3 - 12y^2 - 36y + 2$

$$p = \frac{\partial f}{\partial x} = 2x,$$

$$q = \frac{\partial f}{\partial y} = 12y^2 - 24y - 36$$

$$r = \frac{\partial^2 f}{\partial x^2} = 2, \quad s = \frac{\partial^2 f}{\partial x \partial y} = 0, \quad t = \frac{\partial^2 f}{\partial y^2} = 24y - 24 = 24(y - 1)$$

$$p = \frac{\partial f}{\partial x} = 0 \Rightarrow 2x = 0 \Rightarrow \mathbf{x = 0} \dots \dots \dots (1)$$

$$q = \frac{\partial f}{\partial y} = 0 \Rightarrow 12y^2 - 24y - 36 = 0 \Rightarrow \mathbf{y^2 - 2y - 3 = 0} \dots \dots \dots (2)$$

From (2) we get, $y = -1$ or $y = 3$

Therefore stationary points are $(0, -1)$ and $(0, 3)$

$$\text{Now } rt - s^2 = 2(24(y - 1)) - (0)^2 = 48(y - 1)$$

Point	$rt - s^2$ $= 48(y - 1)$	$r = 2$	Conclusion
$(0, -1)$	$-96 < 0$	$-$	Neither maxima nor minima at $(0, -1)$. $(0, -1)$ is a saddle point.
$(0, 3)$	$96 > 0$	$2 > 0$	$f(x, y)$ has minimum value at $(0, 3)$. $f_{min} = f(0, 3)$ $= 0 + 4(3)^3 - 12(3)^2 - 36(3) + 2$ $= 108 - 108 - 108 + 2$ $= -106$

Example 8. Find extreme values of $f(x, y) = x^3 + y^3 - 3axy$ where $a > 0$

Solution : $f(x, y) = x^3 + y^3 - 3axy$

$$p = \frac{\partial f}{\partial x} = 3x^2 - 3ay,$$

$$r = \frac{\partial^2 f}{\partial x^2} = 6x,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -3a,$$

$$q = \frac{\partial f}{\partial y} = 3y^2 - 3ax$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6y$$

$$p = 0 \Rightarrow 3x^2 - 3ay = 0 \Rightarrow \mathbf{x^2 = ay} \dots \dots \dots (1)$$

$$q = 0 \Rightarrow 3y^2 - 3ax = 0 \Rightarrow \mathbf{y^2 = ax} \dots \dots \dots (2)$$

$$x^2 = ay \Rightarrow x^4 = a^2 y^2 = a^2 ax = a^3 x$$

$$\Rightarrow x^4 - a^3 x = 0 \Rightarrow x(x^3 - a^3) = 0$$

$$\Rightarrow x(x - a)(x^2 + ax + a^2) = 0$$

$$\Rightarrow x = 0 \text{ or } x - a = 0 \text{ or } x^2 + ax + a^2 = 0$$

Now $x^2 + ax + a^2 = 0$ have complex (imaginary) roots.

$$\Rightarrow x = 0 \text{ or } x - a = 0 \text{ i.e. } x = a$$

$$x = 0 \text{ \& } x^2 = ay \Rightarrow y = 0$$

$$x = a \text{ \& } x^2 = ay \Rightarrow y = a$$

Therefore stationary points are $(0, 0)$ and (a, a)

$$\text{Now } rt - s^2 = (6x)(6y) - (-3a)^2 = 36xy - 9a^2$$

Point	$rt - s^2$ $= 36xy - 9a^2$	$r = 6x$	Conclusion
$(0, 0)$	$-9a^2 < 0$	$-$	Neither maxima nor minima at $(0, 0)$. $(0, 0)$ is a saddle point.
(a, a)	$27a^2 > 0$	$6a > 0$	$f(x, y)$ has minimum value at (a, a) . $f_{min} = a^3 + a^3 - 3a^3 = -a^3$

Example 9. Find extreme values of $f(x, y) = x^4 + y^4 - 4xy + 1$

Solution : $f(x, y) = x^4 + y^4 - 4xy + 1$

$$p = \frac{\partial f}{\partial x} = 4x^3 - 4y,$$

$$q = \frac{\partial f}{\partial y} = 4y^3 - 4x$$

$$r = \frac{\partial^2 f}{\partial x^2} = 12x^2,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -4,$$

$$t = \frac{\partial^2 f}{\partial y^2} = 12y^2$$

$$p = 0 \Rightarrow 4x^3 - 4y = 0 \Rightarrow \mathbf{x^3 = y} \dots \dots \dots (1)$$

$$q = 0 \Rightarrow 4y^3 - 4x = 0 \Rightarrow \mathbf{y^3 = x} \dots \dots \dots (2)$$

$$x^3 = y \Rightarrow (x^3)^3 = y^3 \Rightarrow \mathbf{x^9 = x}$$

$$\Rightarrow x^9 - x = 0 \Rightarrow x(x^8 - 1) = 0$$

$$\Rightarrow x(x^4-1)(x^4+1) = 0$$

$$\Rightarrow x(x^2-1)(x^2+1)(x^4+1) = 0$$

$$\Rightarrow x = 0 \text{ or } x^2 - 1 = 0 \text{ or } x^2 + 1 = 0 \text{ or } x^4 + 1 = 0$$

Now $x^2 + 1 = 0$ and $x^4 + 1 = 0$ have complex (imaginary) roots.

$$\therefore x = 0 \text{ or } x^2 - 1 = 0 \Rightarrow x = 0, \quad x = \pm 1$$

$$x = 0 \text{ \& } x^3 = y \Rightarrow y = 0$$

$$x = 1 \text{ \& } x^3 = y \Rightarrow y = 1$$

$$x = -1 \text{ \& } x^3 = y \Rightarrow y = -1$$

Therefore stationary points are $(0, 0)$, $(1, 1)$ and $(-1, -1)$

$$\begin{aligned} \text{Now } rt - s^2 &= (12x^2)(12y^2) - (-4)^2 \\ &= 144x^2y^2 - 16 \end{aligned}$$

Point	$rt - s^2$ $= 144x^2y^2 - 16$	$r = 12x^2$	Conclusion
$(0, 0)$	$-16 < 0$	$-$	Neither maxima nor minima at $(0, 0)$. $(0, 0)$ is a saddle point.
$(1, 1)$	$128 > 0$	$12 > 0$	f has minimum value at $(1, 1)$. $f_{min} = 1^4 + 1^4 - 4(1)(1) + 1 = -1$
$(-1, -1)$	$128 > 0$	$12 > 0$	f has minimum value at $(-1, -1)$. $f_{min} = (-1)^4 + (-1)^4 - 4(-1)(-1) + 1 = -1$

Example 10. Find extreme values of $f(x, y) = x^3 + y^3 - 3x - 12y + 20$

Solution : $f(x, y) = x^3 + y^3 - 3x - 12y + 20$

$$p = \frac{\partial f}{\partial x} = 3x^2 - 3,$$

$$q = \frac{\partial f}{\partial y} = 3y^2 - 12$$

$$r = \frac{\partial^2 f}{\partial x^2} = 6x,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0,$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6y$$

$$p = 0 \Rightarrow 3x^2 - 3 = 0 \Rightarrow x^2 - 1 = 0 \Rightarrow \mathbf{x = \pm 1} \dots \dots \dots (1)$$

$$q = 0 \Rightarrow 3y^2 - 12 = 0 \Rightarrow y^2 - 4 = 0 \Rightarrow \mathbf{y = \pm 2} \dots \dots \dots (2)$$

Therefore stationary points are $(1, 2)$, $(1, -2)$, $(-1, 2)$ and $(-1, -2)$

$$\text{Now } rt - s^2 = (6x)(6y) - 0 = 36xy$$

$$\Rightarrow x(x^4-1)(x^4+1) = 0$$

$$\Rightarrow x(x^2-1)(x^2+1)(x^4+1) = 0$$

$$\Rightarrow x = 0 \text{ or } x^2 - 1 = 0 \text{ or } x^2 + 1 = 0 \text{ or } x^4 + 1 = 0$$

Now $x^2 + 1 = 0$ and $x^4 + 1 = 0$ have complex (imaginary) roots.

$$\therefore x = 0 \text{ or } x^2 - 1 = 0 \Rightarrow x = 0, \quad x = \pm 1$$

$$x = 0 \text{ \& } x^3 = y \Rightarrow y = 0$$

$$x = 1 \text{ \& } x^3 = y \Rightarrow y = 1$$

$$x = -1 \text{ \& } x^3 = y \Rightarrow y = -1$$

Therefore stationary points are $(0, 0)$, $(1, 1)$ and $(-1, -1)$

$$\begin{aligned} \text{Now } rt - s^2 &= (12x^2)(12y^2) - (-4)^2 \\ &= 144x^2y^2 - 16 \end{aligned}$$

Point	$rt - s^2 = 36xy$	$r = 6x$	Conclusion
$(1, 2)$	$72 > 0$	$6 > 0$	f has minimum value at $(1, 2)$. $f_{min} = 1^3 + 2^3 - 3(1) - 12(2) + 20$ $= 2$
$(1, -2)$	$-72 < 0$	$6 > 0$	Neither maxima nor minima at $(1, -2)$. $(1, -2)$ is a saddle point.
$(-1, 2)$	$-72 < 0$	$-6 < 0$	Neither maxima nor minima at $(-1, 2)$. $(-1, 2)$ is a saddle point.
$(-1, -2)$	$72 > 0$	$-6 < 0$	f has maximum value at $(-1, -2)$. f_{max} $= (-1)^3 + (-2)^3 - 3(-1) - 12(-2) + 20$ $= 48$

Example 11. Find extreme values of $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$

Solution : $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$

$$p = \frac{\partial f}{\partial x} = 3x^2 + 3y^2 - 6x,$$

$$q = \frac{\partial f}{\partial y} = 6xy - 6y$$

$$r = \frac{\partial^2 f}{\partial x^2} = 6x - 6,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 6y,$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6x - 6$$

$$p = 0 \Rightarrow 3x^2 + 3y^2 - 6x = 0 \Rightarrow x^2 + y^2 - 2x = 0 \dots \dots \dots (1)$$

$$q = 0 \Rightarrow 6xy - 6y = 0 \Rightarrow y(x - 1) = 0 \dots \dots \dots (2)$$

From (2), $y = 0$ or $x = 1$

Putting $y = 0$ in (1), we get $x^2 - 2x = 0 \Rightarrow x(x - 2) \Rightarrow x = 0, 2$

Putting $x = 1$ in (1), we get $1 + y^2 - 2 = 0 \Rightarrow y^2 - 1 = 0 \Rightarrow y = \pm 1$

Therefore stationary points are $(0, 0), (2, 0), (1, 1)$ & $(1, -1)$

$$\begin{aligned}
 rt - s^2 &= (6x - 6)(6x - 6) - (6y)^2 = (6x - 6)^2 - 36y^2 \\
 &= 36[(x - 1)^2 - y^2]
 \end{aligned}$$

Point	$rt - s^2$ $= 36[(x - 1)^2 - y^2]$	r $= 6x - 6$	Conclusion
(0, 0)	$36 > 0$	$-6 < 0$	f has maximum value at (0,0). $f_{max} = 4$
(2, 0)	$36 > 0$	$6 > 0$	f has minimum value at (0,0). $f_{min} = (2)^3 + 0 - 3(2)^2 - 0 + 4 = 0$
(1, 1)	$-36 < 0$	0	Neither maxima nor minima at (1, 1). (1, 1) is a saddle point.
(1, -1)	$-36 < 0$	0	Neither maxima nor minima at (1, -1). (1, -1) is a saddle point.

Example 12. Find extreme values of $f(x, y) = xy(a - x - y)$ where $a > 0$

Solution : $p = \frac{\partial f}{\partial x} = ay - 2xy - y^2, \quad q = \frac{\partial f}{\partial y} = ax - x^2 - 2xy$

$$r = \frac{\partial^2 f}{\partial x^2} = -2y, \quad s = \frac{\partial^2 f}{\partial x \partial y} = a - 2x - 2y, \quad t = \frac{\partial^2 f}{\partial y^2} = -2x$$

$$p = 0 \Rightarrow ay - 2xy - y^2 = 0 \Rightarrow \mathbf{y(a - 2x - y) = 0} \dots \dots \dots (1)$$

$$q = 0 \Rightarrow ax - x^2 - 2xy = 0 \Rightarrow \mathbf{x(a - x - 2y) = 0} \dots \dots \dots (2)$$

From(1), $\mathbf{y = 0}$ or $\mathbf{a - 2x - y = 0}$

From(2), $\mathbf{x = 0}$ or $\mathbf{a - x - 2y = 0}$

Considering following four pairs of equations of (1) and (2) , we get

$$y = 0 \text{ and } x = 0 \implies (x, y) = (0, 0)$$

$$y = 0 \text{ and } a - x - 2y = 0 \implies (x, y) = (a, 0)$$

$$a - 2x - y = 0 \text{ and } x = 0 \implies (x, y) = (0, a)$$

$$a - 2x - y = 0 \text{ and } a - x - 2y = 0 \implies (x, y) = \left(\frac{a}{3}, \frac{a}{3}\right)$$

Therefore stationary points are $(0, 0)$, $(a, 0)$, $(0, a)$ and $\left(\frac{a}{3}, \frac{a}{3}\right)$.

$$\begin{aligned} \text{Now } rt - s^2 &= (-2y)(-2x) - (a - 2x - 2y)^2 \\ &= 4xy - (a - 2x - 2y)^2 \end{aligned}$$

Point	$rt - s^2$ $= 4xy - (a - 2x - 2y)^2$	$r = -2y$	Conclusion
$(0, 0)$	$-a^2 < 0$	—	Neither maxima nor minima. $(0, 0)$ is a saddle point.
$(a, 0)$	$-a^2 < 0$	—	Neither maxima nor minima. $(a, 0)$ is a saddle point.
$(0, a)$	$-a^2 < 0$	—	Neither maxima nor minima. $(0, a)$ is a saddle point.
$\left(\frac{a}{3}, \frac{a}{3}\right)$	$4 \frac{a}{3} \frac{a}{3} - \left(a - \frac{2a}{3} - \frac{2a}{3}\right)^2$ $= \frac{a^2}{3} > 0$	$-\frac{2a}{3} < 0$	$f(x, y)$ has maximum value. $f_{max} = \frac{a}{3} \frac{a}{3} \left(a - \frac{a}{3} - \frac{a}{3}\right) = \frac{a^3}{27}$

Example 13. Find extreme values of $f(x, y) = x^3y^2(1 - x - y)$.

Solution : $f(x, y) = x^3y^2 - x^4y^2 - x^3y^3$

$$p = \frac{\partial f}{\partial x} = 3x^2y^2 - 4x^3y^2 - 3x^2y^3,$$

$$q = \frac{\partial f}{\partial y} = 2x^3y - 2x^4y - 3x^3y^2$$

$$r = \frac{\partial^2 f}{\partial x^2} = 6xy^2 - 12x^2y^2 - 6xy^3 = xy^2(6 - 12x - 6y)$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 6x^2y - 8x^3y - 9x^2y^2 = x^2y(6 - 8x - 9y)$$

$$t = \frac{\partial^2 f}{\partial y^2} = 2x^3 - 2x^4 - 6x^3y = x^3(2 - 2x - 6y)$$

$$p = 0 \Rightarrow 3x^2y^2 - 4x^3y^2 - 3x^2y^3 = 0 \Rightarrow x^2y^2(3 - 4x - 3y) = 0$$

$$\Rightarrow x = 0, y = 0, 4x + 3y = 3 \dots (1)$$

$$q = 0 \Rightarrow 2x^3y - 2x^4y - 3x^3y^2 = 0 \Rightarrow x^3y(2 - 2x - 3y) = 0$$

$$\Rightarrow x = 0, y = 0, 2x + 3y = 2 \dots (2)$$

Considering six pairs of equations (1) and (2),

$$x = 0$$

$$y = 0$$

$$x = 0$$

$$2x + 3y = 2$$

$$y = 0$$

$$2x + 3y = 2$$

$$4x + 3y = 3$$

$$x = 0$$

$$4x + 3y = 3$$

$$y = 0$$

$$4x + 3y = 3$$

$$2x + 3y = 2$$

Solving these equations we get following stationary points:

$$(0,0), \left(0, \frac{2}{3}\right), (1,0), (0,1), \left(\frac{3}{4}, 0\right) \text{ \& } \left(\frac{1}{2}, \frac{1}{3}\right)$$

Point	$r = xy^2(6 - 12x - 6y)$	$s = x^2y(6 - 8x - 9y)$	$t = x^3(2 - 2x - 6y)$	$rt - s^2$	Conclusion
$(0, 0)$	0	0	0	0	No conclusion.
$\left(0, \frac{2}{3}\right)$	0	0	0	0	No conclusion.
$(1, 0)$	0	0	0		No conclusion.
$(0, 1)$	0	0	0		No conclusion.
$\left(\frac{3}{4}, 0\right)$	0	0	$\frac{27}{128}$	0	No conclusion.
$\left(\frac{1}{2}, \frac{1}{3}\right)$	$-\frac{1}{9} < 0$	$-\frac{1}{12}$	$-\frac{1}{8}$	$\frac{1}{144} > 0$	$f(x, y)$ has maximum value at $\left(\frac{1}{2}, \frac{1}{3}\right)$. $f_{max} = f\left(\frac{1}{2}, \frac{1}{3}\right) = \frac{1}{432}$

Example 14. Find extreme values of $f(x, y) = \sin x + \sin y + \sin(x + y)$

Solution : $f(x, y) = \sin x + \sin y + \sin(x + y)$

$$p = \frac{\partial f}{\partial x} = \cos x + \cos(x + y)$$

$$q = \frac{\partial f}{\partial y} = \cos y + \cos(x + y)$$

$$r = \frac{\partial^2 f}{\partial x^2} = -\sin x - \sin(x + y)$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -\sin(x + y)$$

$$t = \frac{\partial^2 f}{\partial y^2} = -\sin y - \sin(x + y)$$

$$p = 0 \Rightarrow \cos x + \cos(x + y) = 0 \dots \dots \dots (1)$$

$$q = 0 \Rightarrow \cos y + \cos(x + y) = 0 \dots \dots \dots (2)$$

From (1) and (2),

$$\cos x + \cos(x + y) = \cos y + \cos(x + y)$$

$$\Rightarrow \cos x = \cos y$$

$$\Rightarrow x = y$$

Substituting $y = x$ in any equation (1) or (2) we get,

$$\cos x + \cos(2x) = 0$$

$$\Rightarrow \cos x + 2 \cos^2 x - 1 = 0 \Rightarrow 2 \cos^2 x + \cos x - 1 = 0$$

This is quadratic equation in $\cos x$.

$$\therefore \cos x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-1 \pm \sqrt{1^2 - 4(2)(-1)}}{2 \times 2} = \frac{-1 \pm \sqrt{9}}{4} = \frac{-1 \pm 3}{4}$$

$$\Rightarrow \cos x = -1, \frac{1}{2}$$

$$\cos x = -1 \Rightarrow x = \pi \Rightarrow y = \pi$$

$$\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3} \Rightarrow y = \frac{\pi}{3}$$

Therefore stationary points are (π, π) & $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$.

Point	$r =$ $-\sin x$ $-\sin(x+y)$	$s =$ $-\sin(x+y)$	$t =$ $-\sin y$ $-\sin(x+y)$	$rt - s^2$	Conclusion
(π, π)	0	0	0	0	No conclusion.
$\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$	$-\sin \frac{\pi}{3} - \sin \frac{2\pi}{3}$ $= -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}$ $= -\sqrt{3}$	$-\sin \frac{2\pi}{3}$ $= -\frac{\sqrt{3}}{2}$	$-\sin \frac{\pi}{3} - \sin \frac{2\pi}{3}$ $= -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}$ $= -\sqrt{3}$	$\frac{9}{4} > 0$	f has maximum value at $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$ f_{max} $= \sin \frac{\pi}{3} + \sin \frac{\pi}{3} + \sin \frac{2\pi}{3}$ $= \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$

Example 15. Find extreme values of $f(x, y) = \sin x \sin y \sin(x + y)$

Solution : $f(x, y) = \sin x \sin y \sin(x + y)$

$$p = \frac{\partial f}{\partial x} = \sin y [\cos x \sin(x + y) + \sin x \cos(x + y)] = \sin y \sin(2x + y)$$

$$= \frac{1}{2} [\cos 2x - \cos(2x + 2y)] \quad \because \sin A \sin B = \frac{\cos(A - B) - \cos(A + B)}{2}$$

$$q = \frac{\partial f}{\partial y} = \sin x [\cos y \sin(x + y) + \sin y \cos(x + y)] = \sin x \sin(x + 2y)$$

$$= \frac{1}{2} [\cos 2y - \cos(2x + 2y)]$$

$$r = \frac{\partial^2 f}{\partial x^2} = -\sin 2x + \sin(2x + 2y), \quad s = \frac{\partial^2 f}{\partial x \partial y} = \sin(2x + 2y)$$

$$t = \frac{\partial^2 f}{\partial y^2} = -\sin 2y + \sin(2x + 2y)$$

$$p = 0 \Rightarrow \frac{1}{2} [\cos 2x - \cos(2x + 2y)] = 0 \dots \dots \dots (1)$$

$$q = 0 \Rightarrow \frac{1}{2} [\cos 2y - \cos(2x + 2y)] = 0 \dots \dots \dots (2)$$

From (1) and (2),

$$\begin{aligned} \cos 2x - \cos(2x + 2y) &= \cos 2y - \cos(2x + 2y) \\ \Rightarrow \cos 2x &= \cos 2y \Rightarrow 2x = 2y \Rightarrow \mathbf{x = y} \end{aligned}$$

Substituting $\mathbf{y = x}$ in any equation (1) or (2) we get,

$$\frac{1}{2} [\cos 2x - \cos(4x)] = 0$$

$$\Rightarrow \cos 2x - (2 \cos^2 2x - 1) = 0 \Rightarrow \mathbf{2 \cos^2 2x - \cos 2x - 1 = 0}$$

This is quadratic equation in $\cos 2x$.

$$\therefore \cos 2x = \frac{-(-1) \pm \sqrt{1 - 4(2)(-1)}}{2 \times 2} = \frac{1 \pm \sqrt{9}}{4} = \frac{1 \pm 3}{4} = 1, -\frac{1}{2}$$

$$\cos 2x = 1 \Rightarrow 2x = 0 \Rightarrow x = 0 \Rightarrow y = 0$$

$$\cos 2x = -\frac{1}{2} \Rightarrow 2x = \frac{2\pi}{3} \Rightarrow x = \frac{\pi}{3} \Rightarrow y = \frac{\pi}{3}$$

Therefore stationary points are $(0,0)$ & $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$.

Point	$r =$ $-\sin 2x$ $+ \sin(2x + 2y)$	$s =$ $\sin(2x + 2y)$	$t =$ $-\sin 2y$ $+ \sin(2x + 2y)$	$rt - s^2$	Conclusion
$(0,0)$	0	0	0	0	No conclusion.
$\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$	$-\sin \frac{2\pi}{3} + \sin \frac{4\pi}{3}$ $= -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}$ $= -\sqrt{3}$	$\sin \frac{4\pi}{3}$ $= -\frac{\sqrt{3}}{2}$	$-\sin \frac{2\pi}{3} + \sin \frac{4\pi}{3}$ $= -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}$ $= -\sqrt{3}$	$\frac{9}{4} > 0$	f has maximum value at $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$ $f_{\max} = \sin \frac{\pi}{3} \sin \frac{\pi}{3} \sin \frac{2\pi}{3}$ $= \frac{\sqrt{3}}{2} \frac{\sqrt{3}}{2} \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{8}$

Test Your Knowledge

Q.Find extreme values of the following functions.		Stationary points	Extreme values
1.	$f(x, y) = x^2 + y^2 + 6x + 12$	$(-3, 0)$	$f_{\min} = 3$ at $(-3, 0)$
2.	$f(x, y) = 3x^2 - y^2 + x^3$	$(0, 0), (-2, 0)$	$f_{\max} = 4$ at $(-2, 0)$
3.	$f(x, y) = xy + a^3 \left(\frac{1}{x} + \frac{1}{y} \right)$	(a, a)	$f_{\min} = 3a^2$ at (a, a)
4.	$f(x, y) = x^3 + xy^2 + 21x - 12x^2 - 2y^2$	$(1, 0), (7, 0),$ $(2, \pm\sqrt{15})$	$f_{\max} = 10$ at $(1, 0)$ $f_{\min} = -98$ at $(7, 0)$
5.	$f(x, y) = x^3 + y^3 - 63(x + y) + 12xy$	$(-7, -7), (3, 3),$ $(-1, 5), (5, -1)$	$f_{\max} = 784$ at $(-7, -7)$ $f_{\min} = -216$ at $(3, 3)$
6.	$f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$	$(5, 1), (5, -1),$ $(4, 0), (6, 0)$	$f_{\max} = 112$ at $(4, 0)$ $f_{\min} = 108$ at $(6, 0)$
7.	$f(x, y) = 2(x^2 - y^2) - x^4 + y^4$	$(\pm 1, 0), (0, \pm 1)$	$f_{\max} = 1$ at $(\pm 1, 0)$ $f_{\min} = -1$ at $(0, \pm 1)$
8.	$f(x, y) = 120x + 120y - xy - x^2 - y^2$	$(40, 40)$	$f_{\max} = 4800$ at $(40, 40)$

Q. Find extreme values of the functions.		Stationary points	Extreme values
9.	$f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$	$(0,0)$ $(\sqrt{2}, -\sqrt{2})$ $(-\sqrt{2}, \sqrt{2})$	$f_{\min} = -8$ at $(\sqrt{2}, -\sqrt{2})$ & $(-\sqrt{2}, \sqrt{2})$
10.	$f(x, y) = 2(x^2 - y^2) - x^4 + y^4$	$(\pm 1, 0)$ $(0, \pm 1)$	$f_{\max} = 1$ at $(\pm 1, 0)$ $f_{\min} = -1$ at $(0, \pm 1)$
11.	$f(x, y) = x^3 + 3x^2 + y^2 + 4xy.$	$(0,0)$ $\left(\frac{2}{3}, -\frac{4}{3}\right)$	$f_{\min} = 0$ at both points
12.	$f(x, y) = 5x^2 + 10y^2 + 12xy - 30y - 20x + 25$	$\left(\frac{10}{14}, \frac{15}{14}\right)$	$f_{\max} = \frac{350}{144}$
13.	$f(x, y) = x^2 + y^2 + \frac{2}{xy}$	$(1,1)$ $(-1,-1)$	$f_{\min} = 4$ at both points
14.	$f(x, y) = xy(a^2 - x^2 - y^2)$	$(0,0)$ $\left(\frac{a}{2}, \frac{a}{2}\right)$ $\left(-\frac{a}{2}, -\frac{a}{2}\right)$	$f_{\max} = \frac{a^4}{8}$ at $\left(\frac{a}{2}, \frac{a}{2}\right)$ & $\left(-\frac{a}{2}, -\frac{a}{2}\right)$

Q. Find extreme values of the following		Stationary points	Extreme values
15.	$f(x, y) = x^4 + y^4 - x^2 - y^2 + 1$	$(0,0), \left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right),$ $\left(\pm \frac{1}{\sqrt{2}}, 0\right), \left(0, \pm \frac{1}{\sqrt{2}}\right)$	$f_{max} = 1$ at $(0,0)$ $f_{min} = \frac{1}{2}$ at $\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right)$
16.	$f(x, y) = (x^2 + y^2)e^{6x+2x^2}$	$(0,0), (-1,0), \left(-\frac{1}{2}, 0\right)$	$f_{min} = 0$ at $(0,0)$ $f_{min} = e^{-4}$ at $(-1,0)$
17.	$f(x, y) = x^2 + xy + y^2 + \frac{1}{x} + \frac{1}{y}$	$\left(\left(\frac{1}{3}\right)^{\frac{1}{3}}, \left(\frac{1}{3}\right)^{\frac{1}{3}}\right)$	$f_{min} = (3)^{\frac{4}{3}}$
18.	$f(x, y) = xy(3x + 2y + 1)$	$(0,0), \left(-\frac{1}{3}, 0\right),$ $\left(0, -\frac{1}{2}\right), \left(-\frac{1}{9}, -\frac{1}{6}\right)$	$f_{max} = -\frac{5}{162}$ at $\left(-\frac{1}{9}, -\frac{1}{6}\right)$

2. Lagrange's Method of Undermined Multiplier

Lagrange's Method of Undermined Multiplier is very convenient method for finding maximum or minimum values of function of several variables which are connected by given relations.

Working Rule:

Suppose we wish to find the values of x, y and z , for which $f(x, y, z)$ is maximum or minimum

1. Let $f(x, y, z)$ be a function of three variables x, y and z which are connected by the relation $\phi(x, y, z) = 0$.
2. Write an auxiliary function $F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$ where λ is an undermined multiplier to be determined.

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2. Write an auxiliary function $F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$ where λ is an undermined multiplier to be determined.

3. Find $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$ and $\frac{\partial F}{\partial z}$

4. Eliminate λ by solving the equations $\frac{\partial F}{\partial x} = 0$, $\frac{\partial F}{\partial y} = 0$ and $\frac{\partial F}{\partial z} = 0$

together with $\phi(x, y, z) = 0$.

The values of x, y, z so obtained will give the stationary (maximum or minimum) value of $f(x, y, z)$.

Note: This method is useful in finding stationary values of given function but it cannot determine nature of stationary points i.e. whether maximum or minimum.

Example 1: Find the maximum value of $f(x, y, z) = x^2y^3z^4$ subject to the condition $x + y + z = 5$.

Solution: Let $f(x, y, z) = x^2y^3z^4 \dots \dots \dots (1)$

$$x + y + z = 5 \dots \dots \dots (2)$$

$$\phi(x, y, z) = x + y + z - 5 = 0$$

Consider an auxiliary function, $F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$

where λ is an undermined multiplier to be determined.

$$F(x, y, z) = x^2y^3z^4 + \lambda (x + y + z - 5)$$

$$\frac{\partial F}{\partial x} = 2xy^3z^4 + \lambda$$

$$\frac{\partial F}{\partial y} = 3x^2y^2z^4 + \lambda$$

$$\frac{\partial F}{\partial z} = 4x^2y^3z^3 + \lambda$$

$$\frac{\partial F}{\partial x} = 0 \Rightarrow 2xy^3z^4 + \lambda = 0 \Rightarrow \lambda = -2xy^3z^4 \dots \dots \dots (3)$$

$$\frac{\partial F}{\partial y} = 0 \Rightarrow 3x^2y^2z^4 + \lambda = 0 \Rightarrow \lambda = -3x^2y^2z^4 \dots \dots \dots (4)$$

$$\frac{\partial F}{\partial z} = 0 \Rightarrow 4x^2y^3z^3 + \lambda = 0 \Rightarrow \lambda = -4x^2y^3z^3 \dots \dots \dots (5)$$

From (3), (4) and (5)

$$\begin{aligned} -2xy^3z^4 &= -3x^2y^2z^4 = -4x^2y^3z^3 \\ \Rightarrow 2xy^3z^4 &= 3x^2y^2z^4 = 4x^2y^3z^3 \end{aligned}$$

Dividing all by $x^2y^3z^4$ we get

$$\frac{2}{x} = \frac{3}{y} = \frac{4}{z} \Rightarrow y = \frac{3x}{2} \text{ and } z = 2x$$

$$\therefore x + y + z = 5 \Rightarrow x + \frac{3x}{2} + 2x = 5$$

$$\Rightarrow \frac{9x}{2} = 5 \Rightarrow x = \frac{10}{9}$$

$$\Rightarrow y = \frac{3x}{2} = \frac{3}{2} \frac{10}{9} = \frac{15}{9}, \quad z = 2x = \frac{20}{9}$$

Maximum value of $x^2 y^3 z^4 = \left(\frac{10}{9}\right)^2 \left(\frac{15}{9}\right)^3 \left(\frac{20}{9}\right)^4 = \frac{2^{10} \times 5^9}{3^{15}}$

Example 2: Divide 24 into three parts such that continued product of first, square of second and cube of third may be maximum.

Solution: Let x, y, z be three parts of 24 $\Rightarrow x + y + z = 24$

Continued product of first, square of second and cube of third = xy^2z^3

Let $f(x, y, z) = xy^2z^3$

We have to find maximum value of $f(x, y, z) = xy^2z^3$ subject to the condition

$$\phi(x, y, z) = x + y + z - 24 = 0$$

Consider an auxiliary function, $F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$

$$\begin{aligned} \frac{\partial F}{\partial x} &= y^2z^3 + \lambda, & \frac{\partial F}{\partial y} &= 2xyz^3 + \lambda, & \frac{\partial F}{\partial z} &= xy^2z^2 + \lambda \\ & & & & & = xy^2z^3 + \lambda(x + y + z - 24) \end{aligned}$$

$$\frac{\partial F}{\partial x} = 0 \Rightarrow y^2z^3 + \lambda = 0 \Rightarrow \lambda = -y^2z^3 \dots \dots \dots (1)$$

$$\frac{\partial F}{\partial y} = 0 \Rightarrow 2xyz^3 + \lambda = 0 \Rightarrow \lambda = -2xyz^3 \dots \dots \dots (2)$$

$$\frac{\partial F}{\partial z} = 0 \Rightarrow 3xy^2z^2 + \lambda = 0 \Rightarrow \lambda = -3xy^2z^2 \dots \dots \dots (3)$$

From (1), (2) and (3)

$$-y^2z^3 = -2xyz^3 = -3xy^2z^2 \Rightarrow y^2 z^3 = 2xyz^3 = 3xy^2z^2$$

Dividing all by xy^2z^3 we get,

$$\frac{1}{x} = \frac{2}{y} = \frac{3}{z} \Rightarrow y = 2x \text{ \& } z = 3x$$

$$\therefore x + y + z = 24 \Rightarrow x + 2x + 3x = 24$$

$$\Rightarrow 6x = 24 \Rightarrow x = 4$$

$$\therefore x = 4 \Rightarrow y = 2x = 8, \quad z = 3x = 12$$

So by dividing 24 into three parts namely 4, 8 and 12 continued product of first, square of second and cube of third will be maximum

$$\text{Maximum value of continued product} = xy^2z^3 = 4(8)^2 (12)^3 = 442368$$

Example 3: Sum of three numbers is constant. Prove that their product is maximum when they are equal.

Solution: Let x, y, z be three parts such that $x + y + z = a$

Let $f(x, y, z) = xyz$

We have to find maximum value of $f(x, y, z) = xyz$ subject to the condition

$$\phi(x, y, z) = x + y + z - a = 0$$

Consider an auxiliary function, $F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$
 $= xyz + \lambda (x + y + z - a)$

$$\frac{\partial F}{\partial x} = yz + \lambda, \quad \frac{\partial F}{\partial y} = xz + \lambda, \quad \frac{\partial F}{\partial z} = xy + \lambda$$

$$\frac{\partial F}{\partial x} = 0 \Rightarrow yz + \lambda = 0 \Rightarrow \lambda = -yz \dots \dots \dots (1)$$

$$\frac{\partial F}{\partial y} = 0 \Rightarrow xz + \lambda = 0 \Rightarrow \lambda = -xz \dots \dots \dots (2)$$

$$\frac{\partial F}{\partial z} = 0 \Rightarrow xy + \lambda = 0 \Rightarrow \lambda = -xy \dots \dots \dots (3)$$

From (1), (2) and (3)

$$-yz = -xz = -xy \Rightarrow yz = xz = xy$$

Dividing all by xyz we get

$$\frac{1}{x} = \frac{1}{y} = \frac{1}{z} \Rightarrow x = y = z$$

$$\therefore x + y + z = a \Rightarrow x + x + x = a \Rightarrow 3x = a \Rightarrow x = \frac{a}{3} \Rightarrow x = y = z = \frac{a}{3}$$

So Maximum value of product is obtained when $x = y = z$

$$\text{Maximum value of product} = xyz = \frac{a}{3} \frac{a}{3} \frac{a}{3} = \frac{a^3}{27}$$

Example 4: Find points on the surface $z^2 = xy + 1$ nearest to the origin.

Solution: Let (x, y, z) be any point on the surface $z^2 = xy + 1$.

Distance of the point (x, y, z) from origin is

$$d = \sqrt{(x - 0)^2 + (y - 0)^2 + (z - 0)^2} = \sqrt{x^2 + y^2 + z^2}$$

$$d^2 = x^2 + y^2 + z^2$$

$$\text{Let } f(x, y, z) = x^2 + y^2 + z^2$$

We have to find minimum value of $f(x, y, z) = x^2 + y^2 + z^2$ subject to the condition $\phi(x, y, z) = z^2 - xy - 1 = 0$

$$\begin{aligned} \text{Let } F(x, y, z) &= f(x, y, z) + \lambda \phi(x, y, z) \\ &= x^2 + y^2 + z^2 + \lambda (z^2 - xy - 1) \end{aligned}$$

$$\frac{\partial F}{\partial x} = 2x - \lambda y, \quad \frac{\partial F}{\partial y} = 2y - \lambda x, \quad \frac{\partial F}{\partial z} = 2z + 2\lambda z$$

$$\frac{\partial F}{\partial x} = 0 \Rightarrow 2x - \lambda y = 0 \Rightarrow \lambda = \frac{2x}{y} \dots \dots \dots (1)$$

$$\frac{\partial F}{\partial y} = 0 \Rightarrow 2y - \lambda x = 0 \Rightarrow \lambda = \frac{2y}{x} \dots \dots \dots (2)$$

$$\frac{\partial F}{\partial z} = 0 \Rightarrow 2z + 2\lambda z = 0 \Rightarrow z(1 + \lambda) = 0 \Rightarrow 1 + \lambda = 0 \Rightarrow \lambda = -1$$

Putting value of λ in (1) and (2),

$$\frac{2x}{y} = -1 \Rightarrow 2x + y = 0 \quad \text{and} \quad \frac{2y}{x} = -1 \Rightarrow x + 2y = 0$$

$$2x + y = 0 \quad \text{and} \quad x + 2y = 0 \Rightarrow x = 0, y = 0$$

$$\Rightarrow z^2 = xy + 1 = 0 + 1 = 1$$

$$\Rightarrow z = \pm 1$$

Points of the surface $z^2 = xy + 1$ nearest to origin are $(0,0,1)$ and $(0,0,-1)$

Example 5 : Find minimum distance of the plane $3x + 2y + z = 12$ from origin.

Solution: Let (x, y, z) be any point on the plane $3x + 2y + z = 12$.

Distance of the point (x, y, z) from origin is

$$d = \sqrt{(x - 0)^2 + (y - 0)^2 + (z - 0)^2} = \sqrt{x^2 + y^2 + z^2}$$

$$d^2 = x^2 + y^2 + z^2$$

$$\text{Let } f(x, y, z) = x^2 + y^2 + z^2$$

We have to find minimum value of $f(x, y, z) = x^2 + y^2 + z^2$ subject to the condition $\phi(x, y, z) = 3x + 2y + z - 12 = 0$

$$\begin{aligned} \text{Let } F(x, y, z) &= f(x, y, z) + \lambda \phi(x, y, z) \\ &= x^2 + y^2 + z^2 + \lambda (3x + 2y + z - 12) \end{aligned}$$

$$\frac{\partial F}{\partial x} = 2x + 3\lambda, \quad \frac{\partial F}{\partial y} = 2y + 2\lambda, \quad \frac{\partial F}{\partial z} = 2z + \lambda$$

$$\frac{\partial F}{\partial x} = 0 \Rightarrow 2x + 3\lambda = 0 \Rightarrow \lambda = -\frac{2x}{3} \dots \dots \dots (1)$$

$$\frac{\partial F}{\partial y} = 0 \Rightarrow 2y + 2\lambda = 0 \Rightarrow \lambda = -y \dots \dots \dots (2)$$

$$\frac{\partial F}{\partial z} = 0 \Rightarrow 2z + \lambda = 0 \Rightarrow \lambda = -2z \dots \dots \dots (3)$$

From (1), (2) and (3),

$$-\frac{2x}{3} = -y = -2z \Rightarrow \frac{2x}{3} = y = 2z \Rightarrow y = \frac{2x}{3}, \quad z = \frac{x}{3}$$

$$3x + 2y + z = 12 \Rightarrow 3x + 2\left(\frac{2x}{3}\right) + \frac{x}{3} = 12 \Rightarrow \frac{14x}{3} = 12$$

$$\Rightarrow 14x = 36 \Rightarrow x = \frac{36}{14} = \frac{18}{7}$$

$$\Rightarrow y = \frac{2x}{3} = \frac{2}{3} \times \frac{18}{7} = \frac{12}{7}, \quad z = \frac{x}{3} = \frac{1}{3} \times \frac{18}{7} = \frac{6}{7}$$

Minimum distance of the plane $3x + 2y + z = 12$ from origin is

$$d = \sqrt{x^2 + y^2 + z^2} = \sqrt{\left(\frac{18}{7}\right)^2 + \left(\frac{12}{7}\right)^2 + \left(\frac{6}{7}\right)^2} = \sqrt{\frac{324 + 144 + 36}{49}}$$

$$d = \sqrt{\frac{504}{49}} = \sqrt{\frac{36 \times 14}{49}}$$

$$d = \frac{6\sqrt{14}}{7}$$

Example 6 : In triangle ΔABC , show that value of $\cos A \cos B \cos C$ is maximum when triangle is equilateral.

Solution: Note that $A + B + C = \pi$

Let $f(A, B, C) = \cos A \cos B \cos C$

We have to find maximum value of $f(A, B, C) = \cos A \cos B \cos C$ subject to the condition $\phi(A, B, C) = A + B + C - \pi = 0$

Let $F(A, B, C) = f(A, B, C) + \lambda \phi(A, B, C)$
 $= \cos A \cos B \cos C + \lambda (A + B + C - \pi)$

$$\frac{\partial F}{\partial A} = -\sin A \cos B \cos C + \lambda,$$

$$\frac{\partial F}{\partial B} = -\cos A \sin B \cos C + \lambda$$

$$\frac{\partial F}{\partial C} = -\cos A \cos B \sin C + \lambda$$

$$\frac{\partial F}{\partial A} = 0 \Rightarrow -\sin A \cos B \cos C + \lambda = 0 \Rightarrow \lambda = \sin A \cos B \cos C \dots \dots \dots (1)$$

$$\frac{\partial F}{\partial B} = 0 \Rightarrow -\cos A \sin B \cos C + \lambda = 0 \Rightarrow \lambda = \cos A \sin B \cos C \dots \dots \dots (2)$$

$$\frac{\partial F}{\partial C} = 0 \Rightarrow -\cos A \cos B \sin C + \lambda = 0 \Rightarrow \lambda = \cos A \cos B \sin C \dots \dots \dots (3)$$

From (1), (2) and (3),

$$\sin A \cos B \cos C = \cos A \sin B \cos C = \cos A \cos B \sin C$$

Dividing all by $\cos A \cos B \cos C$ we get,

$$\tan A = \tan B = \tan C$$

$$\Rightarrow A = B = C = \frac{\pi}{3} \Rightarrow \text{Triangle is equilateral}$$

$$\text{Maximum value of } \cos A \cos B \cos C = \cos \frac{\pi}{3} \cos \frac{\pi}{3} \cos \frac{\pi}{3} = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

Example 7 : Show that if the perimeter of triangle is constant then triangle has maximum area when it is equilateral.

Solution: Let x, y and z be the sides of the triangle.

Perimeter of the triangle is, $s = \frac{x + y + z}{2}$

Area of the triangle, $A = \sqrt{s(s - x)(s - y)(s - z)}$
 $A^2 = s(s - x)(s - y)(s - z)$

Let $\phi(x, y, z) = x + y + z - 2s = 0$.

We have to find maximum value of $f(x, y, z) = s(s - x)(s - y)(s - z)$
subject to the condition $\phi(x, y, z) = x + y + z - 2s = 0$

Consider an auxiliary function, $F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$

$F(x, y, z) = s(s - x)(s - y)(s - z) + \lambda(x + y + z - 2s)$

$\frac{\partial F}{\partial x} = -s(s - y)(s - z) + \lambda, \quad \frac{\partial F}{\partial y} = -s(s - x)(s - z) + \lambda$

$$\frac{\partial F}{\partial z} = -s(s-x)(s-y) + \lambda$$

$$\frac{\partial F}{\partial x} = 0 \Rightarrow -s(s-y)(s-z) + \lambda = 0 \Rightarrow \lambda = s(s-y)(s-z) \dots \dots \dots (1)$$

$$\frac{\partial F}{\partial y} = 0 \Rightarrow -s(s-x)(s-z) + \lambda = 0 \Rightarrow \lambda = s(s-x)(s-z) \dots \dots \dots (2)$$

$$\frac{\partial F}{\partial z} = 0 \Rightarrow -s(s-x)(s-y) + \lambda = 0 \Rightarrow \lambda = s(s-x)(s-y) \dots \dots \dots (3)$$

From (1), (2) and (3),

$$s(s-y)(s-z) = s(s-x)(s-z) = s(s-x)(s-y)$$

$$s(s-y)(s-z) = s(s-x)(s-z) \quad \text{and} \quad s(s-x)(s-z) = s(s-x)(s-y)$$

$$\Rightarrow s-y = s-x \quad \text{and} \quad s-z = s-y$$

$$\Rightarrow y = x \quad \text{and} \quad z = y$$

$$\Rightarrow x = y = z \Rightarrow \text{Hence, the triangle is equilateral.}$$

Example 8 : Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.

Solution: Let x, y and z be the length, breadth and height of the rectangular solid. Volume of solid is, $V = xyz$.

Equation of sphere is $x^2 + y^2 + z^2 = a^2$.

Let $f(x, y, z) = xyz$

Let $\phi(x, y, z) = x^2 + y^2 + z^2 - a^2 = 0$.

We have to find maximum value of $f(x, y, z) = xyz$ subject to the condition $\phi(x, y, z) = x^2 + y^2 + z^2 - a^2 = 0$.

Consider an auxiliary function, $F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$

$\therefore F(x, y, z) = xyz + \lambda (x^2 + y^2 + z^2 - a^2)$

$$\frac{\partial F}{\partial x} = yz + 2x\lambda, \quad \frac{\partial F}{\partial y} = xz + 2y\lambda, \quad \frac{\partial F}{\partial z} = xy + 2z\lambda$$

$$\frac{\partial F}{\partial x} = 0 \Rightarrow yz + 2x\lambda = 0 \Rightarrow \lambda = -\frac{yz}{2x} \dots \dots \dots (1)$$

$$\frac{\partial F}{\partial y} = 0 \Rightarrow xz + 2y\lambda = 0 \Rightarrow \lambda = -\frac{xz}{2y} \dots \dots \dots (2)$$

$$\frac{\partial F}{\partial z} = 0 \Rightarrow xy + 2z\lambda = 0 \Rightarrow \lambda = -\frac{xy}{2z} \dots \dots \dots (3)$$

From (1), (2) and (3),

$$-\frac{yz}{2x} = -\frac{xz}{2y} = -\frac{xy}{2z} \Rightarrow \frac{x}{yz} = \frac{y}{xz} = \frac{z}{xy}$$

Multiplying all by xyz we get,

$$\Rightarrow x^2 = y^2 = z^2 \Rightarrow x = y = z$$

Hence, the rectangular solid is a cube.

Test Your Knowledge

1. If $xyz = 8$, find the values of x, y, z for which $f = \frac{5xyz}{x + 2y + 4z}$ is a maximum.
2. Find the maximum value of x^2yz^3 subject to the condition $2x + y + 3z = a$.
3. Find the largest product of the numbers x, y and z when $x + y + z^2 = 16$.
4. Find the largest product of the numbers x, y and z when $x^2 + y^2 + z^2 = 9$.
5. Divide 120 into three parts such that sum of their products taken two at a time shall be maximum.
6. Prove that the stationary value of $x^m y^n z^p$ under the condition $x + y + z = a$ is

$$m^m n^n p^p \left(\frac{a}{m + n + p} \right)^{m+n+p}.$$

7. Decompose a positive number a into three parts so that their product is maximum.
8. Find the minimum value of $x^2 + y^2 + z^2$ when (i) $x + y + z = 3a$
(ii) $xy + yz + zx = 3a^2$ (iii) $xyz = a^3$
9. In $\triangle ABC$, show that value of $\sin A \sin B \sin C$ is minimum when it is equilateral.

10. Find the point on the plane $ax + by + cz = p$ at which the function $f = x^2 + y^2 + z^2$ has a minimum value and find this minimum value of f .

11. If $u = \frac{x^2}{a^3} + \frac{y^2}{b^3} + \frac{z^2}{c^3}$ where $x + y + z = 1$ then prove that the stationary value

of u is given by $x = \frac{a^3}{a^3 + b^3 + c^3}, y = \frac{b^3}{a^3 + b^3 + c^3}, z = \frac{c^3}{a^3 + b^3 + c^3}$

12. Show that the stationary value of $a^3x^2 + b^3y^2 + c^3z^2$, where $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$

is given by $x = \frac{a + b + c}{a}, y = \frac{a + b + c}{b}, z = \frac{a + b + c}{c}$

13. The temperature $T(x, y, z)$ at any point in space is $T = kxyz^2$. Find the highest temperature on surface of the sphere $x^2 + y^2 + z^2 = a^2$.

14. The temperature $u(x, y, z)$ at any point in space is $u = 400xyz^2$. Find the highest temperature on surface of the sphere $x^2 + y^2 + z^2 = 1$. **Ans:** 50

15. Determine the perpendicular distance of the point (a, b, c) from the plane $lx + my + nz = 0$.
16. A space probe in the shape of the ellipsoid $4x^2 + y^2 + 4z^2 = 16$ enters the earth's atmosphere and its surface begins to heat. After 1 hour, the temperature at point (x, y, z) on the surface of the probe $T = 4x^2 + 4yz - 16z + 600$.
Find the hottest points on the probe's surface. **Ans:** $\left(\pm \frac{4}{3}, -\frac{4}{3}, -\frac{4}{3}\right)$
17. A flat circular plate is heated so that the temperature at any point (x, y) is $T(x, y) = x^2 + 2y^2 - x$. Find the coldest point on the plate.
18. Find minimum distance from point $\left(0, 0, \frac{25}{9}\right)$ to the surface $z = xy$. **Ans:** $\frac{\sqrt{41}}{3}$
19. Find a point in the plane $x + 2y + 3z = 13$ nearest to the point $(1, 1, 1)$.
20. Find the shortest distance from origin to the surface $xyz^2 = 2$.
21. Find the minimum and maximum distance from the point $(1, 2, 2)$ to the sphere $x^2 + y^2 + z^2 = 36$.

22. Find the maximum and minimum distances of the point $(3, 4, 12)$ from the unit sphere with centre at origin.
23. Determine the point on the paraboloid $z = x^2 + y^2$ which is closest to the point $(3, -6, 4)$.
24. A rectangular box without a lid (open box) is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box.
25. A rectangular box open at the top is to have a volume of 32 cubic units. Find the dimensions of the box requiring least material for its construction.
26. A closed rectangular box has length twice its breadth and has constant volume 243 m^3 . Determine the dimensions of the box requiring least surface area.
27. Find the dimensions of the rectangular box of maximum capacity whose surface area is 432 m^2 if (i) box is open at the top (ii) box is closed.
28. If the total surface area of a closed rectangular box is 108 sq.cm , find the dimensions of the box having maximum capacity.
29. An aquarium with rectangular sides and bottom (and no top) is to hold 32 litres. Find its dimensions so that it will use the least amount of material.

5.Jacobian

Let $u = f(x, y)$ and $v = g(x, y)$ be two functions of x and y .

Then Jacobian of u, v with respect to x, y is given by $J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$

Similarly if $u = f(x, y, z)$, $v = g(x, y, z)$ and $w = h(x, y, z)$ are three functions of x, y and z . Then Jacobian of u, v, w with respect to x, y, z is given by

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Jacobian of Explicit functions

e.g. 1) $u = x^2 - y^2$ and $v = 2xy$ then

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4x^2 + 4y^2$$

e.g. 2) $x = r \cos \theta$ and $y = r \sin \theta$ then

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

e.g. 3) $u = x \sin y$ and $v = y \sin x$ then

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \sin y & x \cos y \\ y \cos x & \sin x \end{vmatrix} = \sin x \sin y - xy \cos x \cos y$$

e.g. 4) $u = \frac{x + y}{1 - xy}$ and $v = \tan^{-1}x + \tan^{-1}y$ then

$$\frac{\partial u}{\partial x} = \frac{(1 - xy)(1) - (x + y)(-y)}{(1 - xy)^2} = \frac{1 - xy + xy + y^2}{(1 - xy)^2} = \frac{1 + y^2}{(1 - xy)^2}$$

$$\frac{\partial u}{\partial y} = \frac{(1 - xy)(1) - (x + y)(-x)}{(1 - xy)^2} = \frac{1 - xy + xy + x^2}{(1 - xy)^2} = \frac{1 + x^2}{(1 - xy)^2}$$

$$\frac{\partial v}{\partial x} = \frac{1}{1+x^2} \quad \text{and} \quad \frac{\partial v}{\partial y} = \frac{1}{1+y^2}$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1+y^2}{(1-xy)^2} & \frac{1+x^2}{(1-xy)^2} \\ \frac{1}{1+x^2} & \frac{1}{1+y^2} \end{vmatrix}$$

$$= \frac{1+y^2}{(1-xy)^2} \frac{1}{1+y^2} - \frac{1+x^2}{(1-xy)^2} \frac{1}{1+x^2}$$

$$= \frac{1}{(1-xy)^2} - \frac{1}{(1-xy)^2}$$

$$= 0$$

e.g.5) $u = xyz$, $v = x^2 + y^2 + z^2$, $w = x + y + z$ then

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} yz & xz & xy \\ 2x & 2y & 2z \\ 1 & 1 & 1 \end{vmatrix}$$

By $C_2 - C_1$ and $C_3 - C_1$

$$J = 2 \begin{vmatrix} yz & xz - yz & xy - yz \\ x & y - x & z - x \\ 1 & 0 & 0 \end{vmatrix} = 2 \begin{vmatrix} yz & (x - y)z & (x - z)y \\ x & y - x & z - x \\ 1 & 0 & 0 \end{vmatrix}$$

$$J = 2 \begin{vmatrix} yz & (x-y)z & (x-z)y \\ x & y-x & z-x \\ 1 & 0 & 0 \end{vmatrix}$$

$$\begin{aligned} J &= 2(x-y)(x-z) \begin{vmatrix} yz & z & y \\ x & -1 & -1 \\ 1 & 0 & 0 \end{vmatrix} \\ &= -2(x-y)(x-z) \begin{vmatrix} 1 & 0 & 0 \\ x & -1 & -1 \\ yz & z & y \end{vmatrix} \\ &= -2(x-y)(x-z)[-y+z] \\ &= -2(x-y)(y-z)(z-x) \end{aligned}$$

e.g.6) $u = \frac{yz}{x}, v = \frac{zx}{y}$ and $w = \frac{xy}{z}$ then

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} -\frac{yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ \frac{z}{y} & -\frac{zx}{y^2} & \frac{x}{y} \\ \frac{y}{z} & \frac{x}{z} & -\frac{xy}{z^2} \end{vmatrix}$$

$$= \frac{1}{x^2 y^2 z^2} \begin{vmatrix} -yz & zx & xy \\ yz & -zx & xy \\ yz & zx & -xy \end{vmatrix} \quad \text{By taking } \frac{1}{x^2}, \frac{1}{y^2}, \frac{1}{z^2} \text{ common from } R_1, R_2, R_3$$

$$= \frac{yz \cdot zx \cdot xy}{x^2 y^2 z^2} \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} \quad \text{By taking } yz, zx, xy \text{ common from } C_1, C_2, C_3$$

$$= -1(1 - 1) - 1(-1 - 1) + 1(1 + 1) = 0 - (-2) + 2 = 4$$

e.g.7) $x = r \sin\theta \cos\phi$, $y = r \sin\theta \sin\phi$, $z = r \cos\theta$ then

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix} = \begin{vmatrix} \sin\theta \cos\phi & r \cos\theta \cos\phi & -r \sin\theta \sin\phi \\ \sin\theta \sin\phi & r \cos\theta \sin\phi & r \sin\theta \cos\phi \\ \cos\theta & -r \sin\theta & 0 \end{vmatrix}$$

$$= \sin\theta \cos\phi [0 + r^2 \sin^2\theta \cos\phi] - r \cos\theta \cos\phi [0 - r \sin\theta \cos\theta \cos\phi]$$

$$- r \sin\theta \sin\phi [-r \sin^2\theta \sin\phi - r \cos^2\theta \sin\phi]$$

$$= r^2 \sin^3\theta \cos^2\phi + r^2 \sin\theta \cos^2\theta \cos^2\phi + r^2 \sin^3\theta \sin^2\phi + r^2 \sin\theta \cos^2\theta \sin^2\phi$$

$$= r^2 \sin^3\theta + r^2 \sin\theta \cos^2\theta$$

$$= r^2 \sin\theta$$

Test Your Knowledge

Q. Find the Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$ for each of the following functions:

1) $u = x + y$ and $v = x - y$

Ans: $-1/2$

2) $u = x^2$ and $v = y^2$

Ans: $4xy$

3) $u = 3x + 5y$ and $v = 4x - 3y$

Ans: -29

4) $u = x + \frac{y^2}{x}$ and $v = \frac{y^2}{x}$

Ans: $\frac{2y}{x}$

5) $u = \frac{y^2}{x}$ and $v = \frac{x^2}{y}$

Ans: -3

6) $u = \frac{x^2}{2x}$ and $v = \frac{y^2}{x^2 + y^2}$

Ans: $\frac{-y}{2x}$

7) $u = \frac{x - y}{1 + xy}$ and $v = \tan^{-1}x - \tan^{-1}y$

Ans: 0

8) $u = x(1 + y)$ and $v = y(1 + x)$

Ans: $1 + x + y$

9) $u = xy$ and $v = \frac{x + y}{x - y}$

Ans: $\frac{(x - y)^2}{4xy}$

10) $u = \frac{2yz}{x}$, $v = \frac{3zx}{y}$ and $w = \frac{4xy}{z}$

Ans: $\frac{1}{96}$

11) $u = \frac{x}{y - z}$, $v = \frac{y}{z - x}$ and $w = \frac{z}{x - y}$

Ans: 0

12) $u = x(1 - y)$, $v = xy(1 - z)$ and $w = xyz$

Ans: x^2y

13) $u = xyz$, $v = xy + yz + zx$ and $w = x + y + z$

Ans: $(x - y)(y - z)(z - x)$

14) $u = x^2$, $v = \sin y$ and $w = e^{-3z}$

Ans: $-6xe^{-3z} \cos y$

15) $u = \frac{x^2 - y^2}{2}$, $v = xy$ and $w = z$

Ans: $\frac{1}{x^2 + y^2}$

Properties of Jacobian

Property No.1: $JJ' = 1$

1. If $u=f(x,y)$ and $v=g(x,y)$ be two functions of x and y and if $J = \frac{\partial(u, v)}{\partial(x, y)} \neq 0$

then $JJ' = 1$ i. e. $\frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(u, v)} = 1$ where $J' = \frac{\partial(x, y)}{\partial(u, v)}$

2. If $u = f(x, y, z)$, $v = g(x, y, z)$ and $w = h(x, y, z)$ be three functions of x, y and z then

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} \cdot \frac{\partial(x, y, z)}{\partial(u, v, w)} = 1$$

Example : Verify $JJ' = 1$ for the following functions.

1) $x = r \cos \theta, y = r \sin \theta$

2) $u = e^x \sec y, v = e^x \tan y$

3) $u = x(1 - y), v = xy$

4) $u = x^2, v = \sin y, w = e^z$

Solution:

1) $x = r \cos \theta, y = r \sin \theta$

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

Now to find $J' = \frac{\partial(r, \theta)}{\partial(x, y)}$, we need to express r and θ in terms of x and y .

$$x^2 + y^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2$$

$$\Rightarrow \mathbf{r} = \sqrt{x^2 + y^2}$$

$$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2}} = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial r}{\partial y} = \frac{2y}{2\sqrt{x^2 + y^2}} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\text{Also } \frac{y}{x} = \frac{r \sin \theta}{r \cos \theta} = \tan \theta \Rightarrow \mathbf{\theta = \tan^{-1} \left(\frac{y}{x} \right)}$$

$$\frac{\partial \theta}{\partial x} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \left(\frac{1}{x}\right) = \frac{x}{x^2 + y^2}$$

$$\begin{aligned}
 J' = \frac{\partial(r, \theta)}{\partial(x, y)} &= \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{\sqrt{x^2 + y^2}} & \frac{y}{\sqrt{x^2 + y^2}} \\ \frac{-y}{x^2 + y^2} & \frac{x}{x^2 + y^2} \end{vmatrix} \\
 &= \frac{x^2}{(x^2 + y^2)\sqrt{x^2 + y^2}} + \frac{y^2}{(x^2 + y^2)\sqrt{x^2 + y^2}} \\
 &= \frac{x^2 + y^2}{(x^2 + y^2)\sqrt{x^2 + y^2}} \\
 &= \frac{1}{\sqrt{x^2 + y^2}} \\
 &= \frac{1}{\sqrt{r^2}} = \frac{1}{r} \\
 \therefore JJ' &= r \cdot \frac{1}{r} = \mathbf{1}
 \end{aligned}$$

$$2) u = e^x \sec y, v = e^x \tan y$$

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} e^x \sec y & e^x \sec y \tan y \\ e^x \tan y & e^x \sec^2 y \end{vmatrix}$$

$$= e^{2x} \sec^3 y - e^{2x} \sec y \tan^2 y$$

$$= e^{2x} \sec y (\sec^2 y - \tan^2 y)$$

$$= e^{2x} \sec y$$

Now to find $J' = \frac{\partial(x, y)}{\partial(u, v)}$, we need to express x and y in terms of u and v .

$$u^2 - v^2 = e^{2x} \sec^2 y - e^{2x} \tan^2 y = e^{2x} (\sec^2 y - \tan^2 y) = e^{2x}$$

$$u^2 - v^2 = e^{2x} \Rightarrow x = \frac{1}{2} \log(u^2 - v^2)$$

$$x = \frac{1}{2} \log(u^2 - v^2) \Rightarrow \frac{\partial x}{\partial u} = \frac{1}{2} \frac{2u}{u^2 - v^2} = \frac{u}{u^2 - v^2}$$

$$\frac{\partial x}{\partial v} = \frac{1}{2} \frac{-2v}{u^2 - v^2} = \frac{-v}{u^2 - v^2}$$

$$\frac{v}{u} = \frac{e^x \tan y}{e^x \sec y} = \cos y \tan y = \sin y$$

$$\Rightarrow y = \sin^{-1} \left(\frac{v}{u} \right)$$

$$\Rightarrow \frac{\partial y}{\partial u} = \frac{1}{\sqrt{1 - \frac{v^2}{u^2}}} \left(-\frac{v}{u^2} \right) = \frac{-v}{u} \frac{1}{\sqrt{u^2 - v^2}}$$

$$\frac{\partial y}{\partial v} = \frac{1}{\sqrt{1 - \frac{v^2}{u^2}}} \left(\frac{1}{u} \right) = \frac{1}{\sqrt{u^2 - v^2}}$$

$$J' = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{u}{u^2 - v^2} & \frac{-v}{u^2 - v^2} \\ \frac{-v}{u} & \frac{1}{\sqrt{u^2 - v^2}} \end{vmatrix}$$

$$J' = \frac{u}{(u^2 - v^2)\sqrt{u^2 - v^2}} - \frac{v^2}{u(u^2 - v^2)\sqrt{u^2 - v^2}}$$

$$= \frac{1}{(u^2 - v^2)\sqrt{u^2 - v^2}} \left[u - \frac{v^2}{u} \right]$$

$$= \frac{1}{(u^2 - v^2)\sqrt{u^2 - v^2}} \left[\frac{u^2 - v^2}{u} \right]$$

$$J' = \frac{1}{(u^2 - v^2) \sqrt{u^2 - v^2}} \left[\frac{u^2 - v^2}{u} \right]$$

$$= \frac{1}{u \sqrt{u^2 - v^2}}$$

$$= \frac{1}{e^x \sec y \sqrt{e^{2x}}}$$

$$= \frac{1}{e^{2x} \sec y}$$

$$\therefore JJ' = e^{2x} \sec y \cdot \frac{1}{e^{2x} \sec y} = \mathbf{1}$$

$$\mathbf{3)} \quad u = x(1 - y), \quad v = xy$$

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 - y & -x \\ y & x \end{vmatrix} = x(1 - y) + xy = x$$

Now to find $J' = \frac{\partial(x, y)}{\partial(u, v)}$, we need to express x and y in terms of u & v .

$$u = x(1 - y) = x - xy = x - v$$

$$\Rightarrow \mathbf{x = u + v}$$

$$v = xy = (u + v)y \Rightarrow \mathbf{y = \frac{v}{u + v} = 1 - \frac{u}{u + v}}$$

$$J' = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ -v & u \end{vmatrix} \frac{1}{(u+v)^2}$$

$$J' = \frac{u}{(u+v)^2} + \frac{v}{(u+v)^2} = \frac{u+v}{(u+v)^2}$$

$$J' = \frac{1}{u+v} = \frac{1}{x}$$

$$\therefore JJ' = x \cdot \frac{1}{x} = \mathbf{1}$$

$$4) \quad u = x^2, v = \sin y, \quad w = e^z$$

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} 2x & 0 & 0 \\ 0 & \cos y & 0 \\ 0 & 0 & e^z \end{vmatrix} = 2xe^z \cos y$$

Now to find $J' = \frac{\partial(x, y, z)}{\partial(u, v, w)}$, we need to express x, y, z in terms of u, v, w .

$$u = x^2 \Rightarrow x = \sqrt{u}$$

$$v = \sin y \Rightarrow y = \sin^{-1} v$$

$$w = e^z \Rightarrow z = \log w$$

$$J' = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} \frac{1}{2\sqrt{u}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{1-v^2}} & 0 \\ 0 & 0 & \frac{1}{w} \end{vmatrix}$$

$$J' = \frac{1}{2\sqrt{u}} \frac{1}{\sqrt{1-v^2}} \frac{1}{w}$$

$$J' = \frac{1}{2x} \frac{1}{\sqrt{1-\sin^2 y}} \frac{1}{e^z} = \frac{1}{2xe^z \cos y}$$

$$\therefore JJ' = 2xe^z \cos y \cdot \frac{1}{2xe^z \cos y} = \mathbf{1}$$

Test Your Knowledge

Q. Verify $JJ' = 1$ for the following functions.

1. $x = e^u \cos v, y = e^u \sin v$

2. $x + y = 2e^u \cos v, x - y = 2ie^u \sin v$

3. $x = uv, y = \frac{u}{v}$

4. $x = u + v, y = uv$

5. $x = \frac{u^2 + v^2}{x}, y = \frac{v^2}{u}$

6. $x = uv, y = \frac{u + v}{u - v}$

7. $x = u \cos v, y = u \sin v$

8. $x = \sin u \cos v, y = \sin u \sin v$

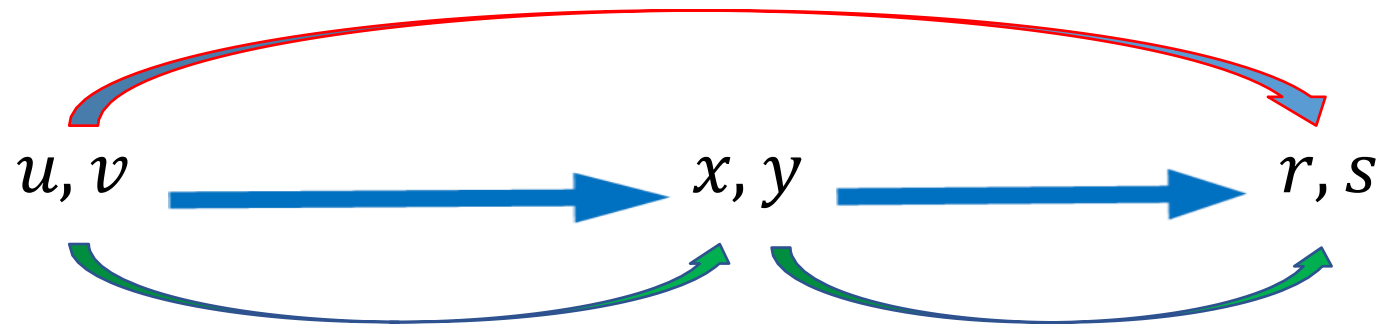
9. $x = e^y \sec u, y = e^y \tan u$

10. $x = u, y = u \tan v, z = w$

11. $x = v^2 + w^2, y = w^2 + u^2, z = u^2 + v^2$

Property No.2: **Jacobian of Composite Functions**

If $u = f(x, y)$ & $v = g(x, y)$ be two functions of x & y and if $x = \phi(r, s)$ & $y = \psi(r, s)$ then u & v are called composite functions of r & s .



$$\therefore \frac{\partial(u, v)}{\partial(r, s)} = \frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(r, s)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial s} \end{vmatrix}$$

If $u = f(x, y, z)$, $v = g(x, y, z)$ & $w = h(x, y, z)$ be three functions of x, y & z and if $x = \phi(r, s, t)$, $y = \psi(r, s, t)$ & $z = \chi(r, s, t)$ then u, v & w are called composite functions of r, s & t .

$$\begin{aligned} \therefore \frac{\partial(u, v, w)}{\partial(r, s, t)} &= \frac{\partial(u, v, w)}{\partial(x, y, z)} \cdot \frac{\partial(x, y, z)}{\partial(r, s, t)} \\ &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial z} \\ \frac{\partial u}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial z} \\ \frac{\partial u}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial s} & \frac{\partial z}{\partial t} \end{vmatrix} \end{aligned}$$

Example: Find Jacobian of the following composite functions.

1. $u = x^2 + y^2, v = 2xy$ and $x = r \cos \theta, y = r \sin \theta$

2. $x = a(u + v), y = b(u - v)$ and $u = r^2 \cos 2\theta, v = r^2 \sin 2\theta$

3. $u = \sqrt{yz}, v = \sqrt{xz}, w = \sqrt{xy}$ and $x = r \sin \theta \cos \phi,$
 $y = r \sin \theta \sin \phi, z = r \cos \theta$

Solution:

1) $u = x^2 + y^2, v = 2xy$ and $x = r \cos \theta, y = r \sin \theta$

Here u & v are composite functions of r & θ .

$$\therefore \frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$\begin{aligned}
\frac{\partial(u, v)}{\partial(r, \theta)} &= \begin{vmatrix} 2x & 2y \\ 2y & 2x \end{vmatrix} \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = (4x^2 - 4y^2)(r \cos^2 \theta + r \sin^2 \theta) \\
&= 4(r^2 \cos^2 \theta - r^2 \sin^2 \theta)(r) \\
&= 4r^3(\cos^2 \theta - \sin^2 \theta) \\
&= 4r^3 \cos 2\theta
\end{aligned}$$

2) $x = a(u + v), y = b(u - v)$ and $u = r^2 \cos 2\theta, v = r^2 \sin 2\theta$

Here x & y are composite functions of r & θ .

$$\begin{aligned}
\frac{\partial(x, y)}{\partial(r, \theta)} &= \frac{\partial(x, y)}{\partial(u, v)} \cdot \frac{\partial(u, v)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial \theta} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial \theta} \end{vmatrix} \\
&= \begin{vmatrix} a & a \\ b & -b \end{vmatrix} \begin{vmatrix} 2r \cos 2\theta & -2r^2 \sin 2\theta \\ 2r \sin 2\theta & 2r^2 \cos 2\theta \end{vmatrix}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial(x, y)}{\partial(u, v)} &= (-ab - ab)(4r^3 \cos^2 \theta + 4r^3 \sin^2 \theta) \\
&= (-2ab)(4r^3) \\
&= -8abr^3
\end{aligned}$$

3) $u = \sqrt{yz}, v = \sqrt{xz}, w = \sqrt{xy}$ and

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

Here u, v & w are composite functions of r, θ & ϕ .

$$\therefore \frac{\partial(u, v, w)}{\partial(r, \theta, \phi)} = \frac{\partial(u, v, w)}{\partial(x, y, z)} \cdot \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial z} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial x} & \frac{\partial w}{\partial z} \\ \frac{\partial u}{\partial z} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial x} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix}$$

$$\begin{aligned}
\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial z} \\ \frac{\partial u}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial z} \\ \frac{\partial u}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} &= \begin{vmatrix} 0 & \frac{1}{2} \sqrt{\frac{z}{y}} & \frac{1}{2} \sqrt{\frac{y}{z}} \\ \frac{1}{2} \sqrt{\frac{z}{x}} & 0 & \frac{1}{2} \sqrt{\frac{x}{z}} \\ \frac{1}{2} \sqrt{\frac{y}{x}} & \frac{1}{2} \sqrt{\frac{x}{y}} & 0 \end{vmatrix} = \frac{1}{8} \frac{1}{\sqrt{xyz}} \begin{vmatrix} 0 & \sqrt{z} & \sqrt{y} \\ \sqrt{z} & 0 & \sqrt{x} \\ \sqrt{y} & \sqrt{x} & 0 \end{vmatrix} \\
&= \frac{1}{8} \frac{1}{\sqrt{xyz}} [0 - \sqrt{z} (0 - \sqrt{xy}) + \sqrt{y} (\sqrt{xz} - 0)] \\
&= \frac{1}{8} \frac{1}{\sqrt{xyz}} [\sqrt{xyz} + \sqrt{xyz}] \\
&= \frac{1}{4}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix} = \begin{vmatrix} \sin\theta \cos\phi & r \cos\theta \cos\phi & -r \sin\theta \sin\phi \\ \sin\theta \sin\phi & r \cos\theta \sin\phi & r \sin\theta \cos\phi \\ \cos\theta & -r \sin\theta & 0 \end{vmatrix} \\
&= \sin\theta \cos\phi [0 + r^2 \sin^2\theta \cos\phi] - r \cos\theta \cos\phi [0 - r \sin\theta \cos\theta \cos\phi] \\
&\quad - r \sin\theta \sin\phi [-r \sin^2\theta \sin\phi - r \cos^2\theta \sin\phi] \\
&= r^2 \sin^3\theta \cos^2\phi + r^2 \sin\theta \cos^2\theta \cos^2\phi + r^2 \sin^3\theta \sin^2\phi + r^2 \sin\theta \cos^2\theta \sin^2\phi \\
&= r^2 \sin^3\theta + r^2 \sin\theta \cos^2\theta \\
&= r^2 \sin\theta \\
\therefore \frac{\partial(u, v, w)}{\partial(r, \theta, \phi)} &= \frac{\partial(u, v, w)}{\partial(x, y, z)} \cdot \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \frac{1}{4} r^2 \sin\theta
\end{aligned}$$

Test Your Knowledge

Q. Find Jacobian of the following composite functions.

1. $u = 2xy, \quad v = x^2 - y^2$ and $x = r \cos \theta, \quad y = r \sin \theta$

Ans: $\frac{\partial(u, v)}{\partial(r, \theta)} = -4r^3$

2. $u = x^2y, \quad v = xy^2$ and $x = r^2 - s^2, \quad y = rs$

Ans: $\frac{\partial(u, v)}{\partial(r, s)} = 6r^2s^2(r^2 - s^2)^3$

3. $u = e^x \cos y, \quad v = e^x \sin y$ and $x = r + s, \quad y = r - s$

Ans: $\frac{\partial(u, v)}{\partial(r, s)} = -2e^{2x}$

4. $u = x^2 - 2y^2, \quad v = 2x^2 - y^2$ and $x = r \cos \theta, \quad y = r \sin \theta$

Ans: $\frac{\partial(u, v)}{\partial(r, \theta)} = 6r^3 \sin 2\theta$

6. Errors and Approximations

Suppose $f(x, y)$ is a continuous function in x and y . If δx and δy are increments (small changes) in x and y then increment δf in f is given by

$$\delta f = f(x + \delta x, y + \delta y) - f(x, y)$$

1. δx is called **absolute error** in x

2. $\frac{\delta x}{x}$ is called **relative error** in x

3. $\frac{\delta x}{x} \times 100$ is called **percentage error** in x

Note: If $u = f(x, y)$ then $\delta u = \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y$

Note: If $u = f(x, y, z)$ then $\delta u = \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y + \frac{\partial u}{\partial z} \delta z$

Example 1: Find the percentage error in calculating the area of a rectangle when an error of 3% is made in measuring each of its sides.

Solution : Let a and b be the sides of the rectangle and A is its area.

$$A = ab$$

$$\log A = \log a + \log b$$

Differentiating we get , $\frac{1}{A} \delta A = \frac{1}{a} \delta a + \frac{1}{b} \delta b$

$$\Rightarrow \frac{\delta A}{A} \times 100 = \left(\frac{\delta a}{a} \times 100 \right) + \left(\frac{\delta b}{b} \times 100 \right)$$

Given: $\frac{\delta a}{a} \times 100 = 3\%$ and $\frac{\delta b}{b} \times 100 = 3\%$

$$\frac{\delta A}{A} \times 100 = 3\% + 3\% = 6\%$$

$\therefore$ Percentage error in calculating the area is 6%

Example 2: Find the percentage error in the area of an ellipse when errors of 2% and 3% are made in measuring its major and minor axes respectively.

Solution : Let $2a$ and $2b$ be the major & minor axes of the ellipse and A is its area.

$$A = \pi ab$$

$$\log A = \log \pi + \log a + \log b$$

Differentiating we get , $\frac{1}{A} \delta A = 0 + \frac{1}{a} \delta a + \frac{1}{b} \delta b$

$$\Rightarrow \frac{\delta A}{A} \times 100 = \left(\frac{\delta a}{a} \times 100 \right) + \left(\frac{\delta b}{b} \times 100 \right)$$

Given: $\frac{\delta a}{a} \times 100 = 2\%$ and $\frac{\delta b}{b} \times 100 = 3\%$

$$\frac{\delta A}{A} \times 100 = 2\% + 3\% = 5\%$$

$\therefore$ Percentage error in calculating the area is 5%

Example 3: Find the percentage error in computing parallel resistance r of two resistances r_1 & r_2 from the formula $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2}$ where r_1 & r_2 both are in error by 2% each.

Solution : We have $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2}$

On differentiating we get, $-\frac{1}{r^2} \delta r = -\frac{1}{r_1^2} \delta r_1 - \frac{1}{r_2^2} \delta r_2$

$$\frac{1}{r} \left(\frac{1}{r} \delta r \right) = \frac{1}{r_1} \left(\frac{1}{r_1} \delta r_1 \right) + \frac{1}{r_2} \left(\frac{1}{r_2} \delta r_2 \right)$$

$$\Rightarrow \frac{1}{r} \left(\frac{\delta r}{r} \times 100 \right) = \frac{1}{r_1} \left(\frac{\delta r_1}{r_1} \times 100 \right) + \frac{1}{r_2} \left(\frac{\delta r_2}{r_2} \times 100 \right)$$

Given: $\frac{\delta r_1}{r_1} \times 100 = 2\%$ and $\frac{\delta r_2}{r_2} \times 100 = 2\%$

$$\therefore \frac{1}{r} \left(\frac{\delta r}{r} \times 100 \right) = \frac{1}{r_1} (2) + \frac{1}{r_2} (2)$$

$$\Rightarrow \frac{1}{r} \left(\frac{\delta r}{r} \times 100 \right) = 2 \left(\frac{1}{r_1} + \frac{1}{r_2} \right) = \frac{2}{r}$$

$$\Rightarrow \frac{\delta r}{r} \times 100 = 2$$

Percentage error in resistance r is also 2%

Example 4: The focal length f of a mirror is found from the formula

$\frac{2}{f} = \frac{1}{v} - \frac{1}{u}$. Find percentage error in f if u and v both of error by 2% each.

Solution : We have $\frac{2}{f} = \frac{1}{v} - \frac{1}{u}$

On differentiating we get, $-\frac{2}{f^2} \delta f = -\frac{1}{v^2} \delta v + \frac{1}{u^2} \delta u$

$$-\frac{2}{f} \left(\frac{1}{f} \delta f \right) = -\frac{1}{v} \left(\frac{1}{v} \delta v \right) + \frac{1}{u} \left(\frac{1}{u} \delta u \right)$$

$$-\frac{2}{f} \left(\frac{\delta f}{f} \times 100 \right) = -\frac{1}{v} \left(\frac{\delta v}{v} \times 100 \right) + \frac{1}{u} \left(\frac{\delta u}{u} \times 100 \right)$$

Given: $\frac{\delta v}{v} \times 100 = 2\%$ and $\frac{\delta u}{u} \times 100 = 2\%$

$$\therefore -\frac{2}{f} \left(\frac{\delta f}{f} \times 100 \right) = -\frac{1}{v} (2) + \frac{1}{u} (2)$$

$$\Rightarrow \frac{1}{f} \left(\frac{\delta f}{f} \times 100 \right) = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{f} \left(\frac{\delta f}{f} \times 100 \right) = \frac{2}{f}$$

$$\Rightarrow \frac{\delta f}{f} \times 100 = 2$$

Percentage error in focal length f is also 2%

Example 5: The resonant frequency in a series electrical circuit is given by $f = \frac{1}{2\pi\sqrt{LC}}$. If the measurement of L and C are in error by 2% and -1% respectively, find the percentage error in f .

Solution: $f = \frac{1}{2\pi\sqrt{LC}} \Rightarrow \log f = \log \frac{1}{2\pi} + \log L^{-1/2} + \log C^{-1/2}$

$$= -\log 2\pi - \frac{1}{2} \log L - \frac{1}{2} \log C$$

Differentiating we get, $\frac{1}{f} \delta f = 0 - \frac{1}{2} \frac{1}{L} \delta L - \frac{1}{2} \frac{1}{C} \delta C$

$$\frac{\delta f}{f} \times 100 = -\frac{1}{2} \left[\left(\frac{\delta L}{L} \times 100 \right) + \left(\frac{\delta C}{C} \times 100 \right) \right]$$

$$\frac{\delta f}{f} \times 100 = -\frac{1}{2} [2 - 1] = -0.5\%$$

Hence percentage error in $f = -0.5\%$

Example 6: Area of a triangle is $\Delta = \frac{1}{2}bc \sin A$. Find percentage error in area if percentage errors in b, c and A are 1%, 2% and 3%. (Given $A = \frac{\pi}{4}$)

Solution : $\Delta = \frac{1}{2}bc \sin A$

Taking logarithm on both sides, we get,

$$\log \Delta = \log \left(\frac{1}{2} \right) + \log b + \log c + \log \sin A$$

Differentiating we get, $\frac{1}{\Delta} \delta \Delta = 0 + \frac{1}{b} \delta b + \frac{1}{c} \delta c + \frac{\cos A}{\sin A} \delta A$

$$\frac{\delta \Delta}{\Delta} \times 100 = \left(\frac{\delta b}{b} \times 100 \right) + \left(\frac{\delta c}{c} \times 100 \right) + \cot A (\delta A \times 100)$$

Given: $\frac{\delta b}{b} \times 100 = 1\%$, $\frac{\delta c}{c} \times 100 = 2\%$ and $\frac{\delta A}{A} \times 100 = 3\%$

$$\frac{\delta \Delta}{\Delta} \times 100 = \left(\frac{\delta b}{b} \times 100 \right) + \left(\frac{\delta c}{c} \times 100 \right) + \cot A (\delta A \times 100)$$

$$\Rightarrow \frac{\delta \Delta}{\Delta} \times 100 = 1 + 2 + \cot A (3 \times A)$$

$$\Rightarrow \frac{\delta \Delta}{\Delta} \times 100 = 3 + \frac{3\pi}{4} \cot \frac{\pi}{4}$$

Percentage error in area Δ is $\left(3 + \frac{3\pi}{4} \right) \%$

Example 7: In calculating the volume of a right circular cone, errors of 2% and 1% are made in height and radius of base respectively. Find the percentage error in calculating the volume.

Solution: Let r be the radius of base, h height & V volume of a right circular cone.

$$V = \frac{1}{3}\pi r^2 h$$

$$\Rightarrow \log V = \log \frac{\pi}{3} + 2 \log r + \log h$$

Differentiating we get, $\frac{1}{V} \delta V = 0 + \frac{2}{r} \delta r + \frac{1}{h} \delta h$

$$\frac{\delta V}{V} \times 100 = 2 \left(\frac{\delta r}{r} \times 100 \right) + \left(\frac{\delta h}{h} \times 100 \right)$$

$$\frac{\delta V}{V} \times 100 = 2(1\%) + 2\% = 4\%$$

Hence, percentage error in volume = 4%.

Example 8: The diameter and the altitude of a can in the shape of a right circular cylinder are measured as 4 cm and 6 cm respectively. The possible error in each measurement is 0.1 cm. Find approximately the possible error in the values computed for volume and lateral surface area.

Solution: Let d and h are diameter and height of the right circular cylinder respectively and V be its volume.

$$V = \pi r^2 h = \pi \left(\frac{d}{2}\right)^2 h = \frac{\pi}{4} d^2 h$$

$$\Rightarrow \log V = \log \frac{\pi}{4} + 2 \log d + \log h$$

Differentiating we get,

$$\frac{1}{V} \delta V = 0 + \frac{2}{d} \delta d + \frac{1}{h} \delta h = \frac{2}{d} \delta d + \frac{1}{h} \delta h$$

$$\therefore \delta V = V \left(\frac{2}{d} \delta d + \frac{1}{h} \delta h \right)$$

Given : $d = 4 \text{ cm}$, $h = 6 \text{ cm}$, $\delta d = 0.1 \text{ cm}$ & $\delta h = 0.1 \text{ cm}$

$$\therefore V = \frac{\pi}{4} d^2 h = \frac{\pi}{4} \times 4^2 \times 6 = 75.36 \text{ cm}^3$$

$$\begin{aligned}\therefore \delta V &= (75.36) \left(\frac{2}{4} (0.1) + \frac{1}{6} (0.1) \right) \\ &= (75.36)(0.067) \\ &= 5.05 \text{ cm}^3\end{aligned}$$

Hence, error in volume = 5.05 cm^3 .

Lateral surface area is, $S = 2\pi rh = \pi dh$

$$\begin{aligned}\Rightarrow \log S &= \log 2\pi + \log d + \log h \\ \Rightarrow \frac{1}{S} \delta S &= 0 + \frac{1}{d} \delta d + \frac{1}{h} \delta h = \frac{1}{d} \delta d + \frac{1}{h} \delta h \\ \Rightarrow \delta S &= S \left(\frac{1}{d} \delta d + \frac{1}{h} \delta h \right)\end{aligned}$$

Given : $d = 4 \text{ cm}$, $h = 6 \text{ cm}$, $\delta d = 0.1 \text{ cm}$ & $\delta h = 0.1 \text{ cm}$

$$\therefore S = \pi dh = \pi \times 4 \times 6 = 75.36 \text{ cm}^2$$

$$\delta S = S \left(\frac{1}{d} \delta d + \frac{1}{h} \delta h \right)$$

$$\begin{aligned} \therefore \delta S &= (75.36) \left(\frac{1}{4} (0.1) + \frac{1}{6} (0.1) \right) \\ &= (75.36)(0.0416) \\ &= 3.14 \text{ cm}^2 \end{aligned}$$

Hence, error in lateral surface area = 3.14 cm^2 .

Example 9: If $D = \frac{a^2}{b} + \frac{c^2}{2}$, Find the percentage error in D if error in measuring a is 0.5% and in measuring b and c are 1% each.

Solution: $D = \frac{a^2}{b} + \frac{c^2}{2}$

Differentiating we get,
$$\delta D = \frac{2a}{b} \delta a - \frac{a^2}{b^2} \delta b + \frac{2c}{2} \delta c$$

$$\frac{\delta D}{D} \times 100 = \frac{100}{D} \left[\frac{2a}{b} \delta a - \frac{a^2}{b^2} \delta b + c \delta c \right]$$

$$\frac{\delta D}{D} \times 100 = \frac{1}{D} \left[\frac{2a^2}{b} \left(\frac{\delta a}{a} \times 100 \right) - \frac{a^2}{b} \left(\frac{\delta b}{b} \times 100 \right) + c^2 \left(\frac{\delta c}{c} \times 100 \right) \right]$$

Given : $\frac{\delta a}{a} \times 100 = 0.5, \quad \frac{\delta b}{b} \times 100 = 1, \quad \frac{\delta c}{c} \times 100 = 1$

$$\frac{\delta D}{D} \times 100 = \frac{1}{D} \left[\frac{2a^2}{b} (0.5) - \frac{a^2}{b} (1) + c^2 (1) \right]$$

$$\frac{\delta D}{D} \times 100 = \frac{c^2}{D}$$

$$= \frac{c^2}{\frac{a^2}{b} + \frac{c^2}{2}}$$

$$= \frac{2bc^2}{2a^2 + 2c^2}$$

Hence, percentage error in $D = \frac{2bc^2}{2a^2 + 2c^2} \%$

Example 10: If the kinetic energy is calculated by the formula $T = \frac{mv^2}{2}$ and m changes from 49 to 49.5 and v changes from 1600 to 1590, find

a) approximately the change in T

b) percentage error in kinetic energy T

Solution: $T = \frac{mv^2}{2} \Rightarrow \log T = \log m + 2 \log v - \log 2$

Differentiating we get, $\frac{\delta T}{T} = \frac{\delta m}{m} + 2 \frac{\delta v}{v} - 0$

$$\delta T = T \left[\frac{\delta m}{m} + 2 \frac{\delta v}{v} \right]$$

Given : $m = 49, v = 1600$

$$\delta m = 49.5 - 49 = 0.5$$

$$\delta v = 1590 - 1600 = -10$$

$$T = \frac{mv^2}{2} = \frac{49 \times 1600^2}{2} = 6,27,20,000$$

$$\begin{aligned}\delta T &= T \left[\frac{\delta m}{m} + 2 \frac{\delta v}{v} \right] = (6,27,20,000) \left[\frac{0.5}{49} + 2 \frac{(-10)}{1600} \right] \\ &= -144000\end{aligned}$$

Hence, approximate change in $T = -144000$.

Percentage error in T is given by,

$$\frac{\delta T}{T} \times 100 = \frac{-144000}{6,27,20,000} \times 100 = -0.2295$$

Hence, percentage error in $T = -0.2295 \%$

Example 11: The period of a simple pendulum with small oscillations is

$T = 2\pi \sqrt{\frac{l}{g}}$ if T is computed using $l = 8 \text{ m}$, $g = 32 \text{ m/s}^2$ find the % error in T if the true values of l and g are 8.05 m and 32.01 m/s^2 respectively.

Solution: $T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow \log T = \log 2\pi + \frac{1}{2} \log l - \frac{1}{2} \log g$

Differentiating we get $\frac{\delta T}{T} = 0 + \frac{1}{2} \frac{\delta l}{l} - \frac{1}{2} \frac{\delta g}{g}$

$$\delta T = \frac{T}{2} \left[\frac{\delta l}{l} - \frac{\delta g}{g} \right]$$

Given : $l = 8 \text{ m}$, $g = 32 \frac{\text{m}}{\text{s}^2}$, $\delta l = 8.05 - 8 = 0.05 \text{ m}$,

$$\delta g = 32.01 - 32 = 0.01 \text{ m/s}^2$$

$$T = 2\pi \sqrt{\frac{8}{32}} = 3.14s$$

$$\begin{aligned}\delta T &= \frac{T}{2} \left[\frac{\delta l}{l} - \frac{\delta g}{g} \right] \\ &= \frac{(3.14)}{2} \left[\frac{0.05}{8} - \frac{0.01}{32} \right]\end{aligned}$$

$$\delta T = 0.00932$$

Hence, error in $T = 0.00932 \text{ s}$

Example 12: Find approximate value of $[(3.82)^2 + 2(2.1)^3]$

Solution: Let

$$u = (x^2 + 2y^3)^{\frac{1}{5}}$$

$$\delta u = \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y$$

$$\delta u = \frac{2x}{5} (x^2 + 2y^3)^{-\frac{4}{5}} \delta x + \frac{6y}{5} (x^2 + 2y^3)^{-\frac{4}{5}} \delta y$$

Let $x = 4$, $y = 2$,

$$\delta x = 3.82 - 4 = -0.18,$$

$$\delta y = 2.1 - 2 = 0.1$$

$$x^2 + 2y^3 = 4^2 + 2 \times 2^3 = 32$$

$$u = (x^2 + 2y^3)^{\frac{1}{5}} = (32)^{\frac{1}{5}} = 2$$

$$\delta u = \frac{2x}{5} (x^2 + 2y^3)^{-\frac{4}{5}} \delta x + \frac{6y}{5} (x^2 + 2y^3)^{-\frac{4}{5}} \delta y$$

$$\delta u = \frac{2 \times 4}{5} (32)^{-\frac{4}{5}} (-0.18) + \frac{6 \times 2}{5} (32)^{-\frac{4}{5}} (0.1)$$

$$\delta u = 0.012$$

$$\text{Approximate value} = u + \delta u$$

$$= 2 + 0.012$$

$$= 2.012$$

Example 13: Find approximate value of $[(0.98)^2 + (2.01)^2 + (1.94)^2]$

Solution: Let

$$u = \sqrt{x^2 + y^2 + z^2}$$

$$\delta u = \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y + \frac{\partial u}{\partial z} \delta z$$

$$\delta u = \frac{x}{\sqrt{x^2 + y^2 + z^2}} \delta x + \frac{y}{\sqrt{x^2 + y^2 + z^2}} \delta y + \frac{z}{\sqrt{x^2 + y^2 + z^2}} \delta z$$

Let $x = 1$, $y = 2$, $z = 2$,

$$\delta x = 0.98 - 1 = -0.02,$$

$$\delta y = 2.01 - 2 = 0.01,$$

$$\delta z = 1.94 - 2 = -0.06$$

$$u = \sqrt{1^2 + 2^2 + 2^2} = 3$$

$$\delta u = \frac{x}{\sqrt{x^2 + y^2 + z^2}} \delta x + \frac{y}{\sqrt{x^2 + y^2 + z^2}} \delta y + \frac{z}{\sqrt{x^2 + y^2 + z^2}} \delta z$$

$$\delta u = \frac{1}{3}(-0.02) + \frac{1}{3}(0.01) + \frac{1}{3}(-.006)$$

$$\delta u = -0.04$$

$$\text{Approximate value} = u + \delta u$$

$$= 3 - 0.04$$

$$= 2.96$$

Test Your Knowledge

1. In calculating the volume of a right circular cylinder, errors of 2% and 1% are found in measuring height and base radius respectively. Find the percentage error in calculated volume of the cylinder.
2. Find the possible percentage error in computing the parallel resistance R of three resistances R_1, R_2, R_3 if R_1, R_2, R_3 are each in error by 1.2%.
3. The diameter of a right circular cylinder are measured to be 5 inches and 8 inches respectively. If each of these dimensions were in error by 0.1 inch, find the % error in the volume of the cylinder.
4. The torsional rigidity N of a length l of a wire is obtained from the formula $N = \frac{8\pi l}{r^4 t^2}$.
Find the % error in N due to -2% error in t , 2% error in r and 1.5% error in l .
5. Find approximate value of $[(2.92)^3 + (5.87)^3]^{1/3}$
6. Find approximate value of $(1.99)^2 (3.01)^3 (0.98)^{1/10}$

7. The deflection at a centre of a rod of length l and diameter d , supported at its ends and loaded at the centre with a weight w , varies as wl^3d^{-4} .

What is the percentage increase in the deflection corresponding to the percentage increase in w , l and d of 3, 2 and 1 respectively.

8. The quantity Q of water flowing over a triangular notch is given by the formula $Q = CH^{5/2}$ where H is head of water and C is constant .

Find the % error in Q if the error in H is that it is measured 0.198 instead of 0.2.

9. If the Horse power required to propel a steamer varies as the cube of the velocity and the square of length. If there is 3 % increase in velocity and 4 % increase in length, find the % increase in HP.

10. The time of swing t of a pendulum of length l under certain condition is

given by $t = 2\pi \sqrt{\frac{l}{g'}}$ where $g' = g \left(\frac{r}{r+h} \right)^2$. Find % error in t , due to

error p % in h and q % in l where r is constant.